

CORE MATHEMATICS GRADE 10: STATISTICS I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 3.0: Statistics and Probability
Sub-Strand	Sub-Strand 3.1: Statistics I
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Collection of Data – Frequency distribution table – Mean, Mode and Median of grouped and ungrouped data – Representation of Data – Interpretation of Data
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) collect data from real-life sources, b) draw a frequency distribution table for grouped and ungrouped data, c) determine mean, mode and median of grouped and ungrouped data, d) represent data using histograms and frequency polygons, e) interpret data from histograms and frequency polygons, f) promote data collection, organisation and use in informed decision making.
Core Competencies	– Critical thinking and problem solving: the learner uses the histograms and frequency polygons to examine data and make conclusions. – Self-Efficacy: the learner discusses with peers the meaning of the term data and explores data sources. – Digital literacy: the learner uses digital devices to draw histograms and frequency polygons to make conclusions. – Communication and collaboration: learner presents and shares findings from data to different groups of people such as school administrators, parents or local community leaders.
Core Values	– Integrity: the learner collects and records data accurately. – Unity: the learner works in groups and uses collected data to draw histograms and frequency polygons. – Love: the learner is considerate of the other as they discuss in pairs and identify the modal class for grouped data.
Science & Engineering Practices	– Asking Questions and Defining Problems: students ask questions about why pedestrians cross at undesignated points and what data would answer them. – Planning and Carrying Out Investigations: students design and run a tally survey of crossing points and pedestrian behaviour along a road near school. – Analyzing and Interpreting Data: students build frequency tables, compute mean, median and mode, and read histograms and

	frequency polygons. – Using Mathematics and Computational Thinking: students use grouped-data formulas, class widths, frequency density and spreadsheets or calculators. – Constructing Explanations: students use data evidence to explain patterns and defend recommendations. – Engaging in Argument from Evidence: students critique which measure of average best represents the data and justify it. – Developing and Using Models: students revise a class data model (table, graph, summary measures) after each lesson. – Obtaining, Evaluating and Communicating Information: students present findings to the school administration or community leaders.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Decision Making): the learner makes informed decisions from data interpretation from histograms and frequency polygons. – Parental Engagement and Empowerment: the learner seeks support and guidance from parents or guardians to collect, organise and present data on an identified situation in the school or community. – Road Safety (link to the suggested data context): pedestrian crossing point data is used to argue for safer roads.
Career Connections	– Statistician / Data Analyst (Kenya National Bureau of Statistics): collects and analyses census and survey data. – Transport and Road Safety Planner (NTSA, KeNHA): uses crossing and traffic data to place footbridges and zebra crossings. – Agricultural Extension Officer: analyses maize and tea yield data from Kericho and the Rift Valley. – Public Health Officer: uses data to track clinic attendance and disease trends. – Economist, Actuary and Data Scientist: use data to guide policy, insurance and technology (M-Pesa, Safaricom analytics).
Focus for Lessons	Learners investigate a real road-safety problem near their school by collecting, organising, summarising and graphing their own data. They move from raw tallies to frequency tables, measures of central tendency, histograms and frequency polygons, and finally to evidence-based recommendations. Each lesson follows Predict, Observe, Explain, DQB and Model Building so that understanding of the anchoring phenomenon grows cumulatively.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use data from our own road to show where pedestrians are really at risk, and why does one average not tell the whole story? KICD KEY INQUIRY QUESTIONS: 1. What is statistics? 2. How do we represent data? 3. How do we use statistics in day-to-day life?
Anchoring Phenomenon	Along a busy road near the school (for example Thika Road in Nairobi, Mombasa Road, or the Kisumu-Kakamega highway), a county road safety officer says: 'The average distance between pedestrian crossing points is only about 400 m, so the road is well served.' Yet learners see pedestrians, including school children, dashing across at undesignated spots, and a local matatu driver says most near-misses happen in just one stretch. Learners tally the crossing points and pedestrians themselves and find that the mean looks reassuring while the raw data show a few very long gaps. The puzzle: how can an average say the road is safe while the data show danger in particular places, and which statistical tools reveal the true story?
Storyline Thread	Lesson 1: Predict whether the road near school is safe from the officer's claim; discuss the meaning of data, explore data sources (surveys, observation, records) and start the DQB. Lesson 2: Collect and record data on pedestrian crossing points and pedestrians crossing at undesignated spots using tallies; build a frequency distribution table for ungrouped data. Lesson 3: Determine mean, mode and median of ungrouped data; compare them to test the officer's 'average' claim and discuss

	which measure best represents the data. Lesson 4: Group the collected data (for example gaps between crossing points) into appropriate classes; draw a grouped frequency table and identify the modal class in pairs. Lesson 5: Calculate the mean and median of grouped data using class midpoints and cumulative frequency; compare with ungrouped results and revise the class model. Lesson 6: Draw histograms with equal and unequal class widths (using frequency density) from the collected data; observe where the long gaps appear. Lesson 7: Draw frequency polygons by hand and with digital devices or spreadsheets; compare polygons from different roads or groups and make conclusions. Lesson 8: Interpret histograms and polygons, make informed recommendations, present findings to the school administration, parents or community leaders, and complete the final model and DQB review for the project.
Supporting Phenomena	– Two Grade 10 classes in Nakuru both have a mean KCSE-style mock score of 62%, but one class has scores bunched together and the other has a wide spread; which class needs more teacher support? – A maize farmer in Trans-Nzoia records bags harvested per acre over 40 plots; the mean is pulled up by a few exceptionally good plots, so the median tells a different story. – A mandazi vendor at Kisumu bus park records daily sales for a month; grouping the sales into classes reveals the busiest sales band that the raw list hides. – Kenyan marathon runners (such as Eliud Kipchoge's training group in Kaptagat) record weekly distances; a histogram shows the typical training band better than a list of numbers.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (40 minutes (one lesson of 40 minutes; the DQB and initial model work can continue in the first 10 minutes of Lesson 2 if needed)): Is the Road Really Well Served? Data and Where It Comes From

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Launch the Thika Road phenomenon. Learners meet the county Road Safety Officer's claim that the average gap between crossing points is about 400 m, so the road is well served. They compare it with what they see at the school gate and with Mzee Otieno's remark that most near-misses happen in one stretch. They then explain what data and statistics are and choose data sources to test the claim.
Knowledge	By the end of the lesson the learner should be able to: (1) explain the meaning of the term data and of statistics; (2) name and describe three sources of data (surveys, observation, records) and give a Thika Road example of each; (3) state what an undesignated crossing is and why the officer's average of 400 m might not tell the whole story. Learners do no calculations in this lesson.
Skills	Making predictions from partial evidence; asking questions from a puzzling phenomenon; planning an investigation (what to measure, how, and with which source); communicating reasoning in pairs and to the class; drawing a first model of the road.
Attitudes	Integrity (the learner collects and records data accurately, as they plan how to do so); Unity (the learner works in groups to plan the data collection); Love (the learner is considerate of others when discussing). Core competencies: Self-Efficacy (the learner discusses with peers the meaning of the term data and explores data sources); Critical thinking and problem solving. PCIs: Road safety; Life Skills (Decision Making).
Key Inquiry Question	1. What is statistics? 2. How do we represent data? 3. How do we use statistics in day-to-day life? (Lesson 1 focuses on question 1 and opens questions 2 and 3 on the DQB.)
Purpose in Storyline	This lesson launches the storyline. Learners accept that the officer's mean of 400 m is arithmetically true but suspect it may be incomplete. The DQB created here is revisited in every later lesson. The data sources and plan made here set up Lesson 2, where learners count the road using tally marks and frequency tables.
Safety Notes	No learner goes onto Thika Road in this lesson. All work is in the classroom, using only the stated claim and the near-miss log. When the real survey is done later, it must be done with Mwalimu Wanjiru and parent or guardian volunteers on a safe footpath, never on the carriageway, and learners must never cross at undesignated spots to count. Treat pedestrian and near-miss stories sensitively, as some learners or relatives may have been involved in road accidents.

B. LESSON OVERVIEW

Mwalimu Wanjiru opens by reading the county Road Safety Officer Ms Halima Abdi's statement about the 8,000 m stretch of Thika Road from Roysambu stage towards Ruiru: it has 21 designated crossing points and the average distance between them is about 400 m, so the road is well served. Learners then recall what they see at Ruaraka Hills Secondary School's gate each morning, with children dashing across away from zebra crossings and footbridges (undesignated crossings), and hear Mzee Otieno, a matatu driver, say that most of his near-misses happen in just one stretch. Learners predict whether the road is safe and record what they would need to know. In the Observe phase they study the officer's statement, a log of 12 near-misses kept by matatu drivers, and three short source cards (survey, observation, records) to see what data is and where it comes from. In the Explain phase pairs build a working meaning of data and statistics and plan which source would answer each question. The class creates the Driving Question Board, sorting questions under the three KICD key inquiry questions. Finally learners draw a first model of the road that shows where they think risk might be, to be revised in later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calculating the mean Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners meet the phenomenon. Mwalimu Wanjiru reads aloud, as a county notice: 'On the 8,000 m stretch of Thika Road from Roysambu stage towards Ruiru there are 21 designated crossing points (zebra crossings and footbridges). The average distance between crossing points is about 400 m, so the road is well served.' (Ms Halima Abdi, County Road Safety Officer). Learners then picture the school gate, which is at the 4,000 m mark, on a weekday morning at 6:30-7:30 am: matatus, boda bodas, and learners and workers dashing across away from the crossings. Mzee Otieno's remark is read: 'Most of my near-misses happen in just one stretch.' Individually, for 3 minutes, learners write on a Predict Card: (a) Is the road near our school safe for pedestrians? Circle: Safe / Not safe / Safe in some places and not in others. (b) One reason from the officer's claim or from what you have seen. (c) One thing you would need to know to be sure. They then share in pairs (Think-Pair-Share), and the class makes a tally on the board of the three choices.	Read the notice slowly, then say: 'Ms Abdi is not careless or dishonest. Let us treat her statement as something we can test.' Ask: 'Picture our gate at 6:45 in the morning. What do you see?' WAIT TIME 10-15 seconds before taking answers. Cold-call 3 learners for what they see, e.g. 'Achieng, what do you see at our gate?', and then read Mzee Otieno's remark. Distribute Predict Cards and give 3 minutes of silent writing. Move around reading cards; select 2 contrasting predictions without naming the writers. Say: 'Turn to your partner and share your prediction and your reason. Your partner should say whether the reason convinces them.' Then take the hand tally on the three choices and record it on the board. Cold-call 4 pairs for the 'what would you need to know' item and write the answers on the board, unedited. Do NOT say whether anyone is right. Say: 'Today we will not decide who is right. We will work out what data could settle it.' Do not introduce any calculation.	Predict-Observe-Explain with Think-Pair-Share. Committing to a prediction before any instruction exposes learners' existing ideas, in particular that an average of 400 m means every part of the road is equally served. Writing 'what I would need to know' moves learners from opinion to the need for evidence, which motivates the idea of data.	Look for: (1) learners who write 'Safe in some places and not in others' or who mention uneven spacing show early awareness of spread; (2) learners who accept the average as proof of safety hold the misconception that one average describes the whole road; note names and revisit in Lesson 3. (3) Predict Cards that give a reason that draws on evidence (the officer's figure, the gate, Mzee Otieno) rather than just opinion. (4) The 'what I need to know' items that mention counting pedestrians, measuring distances or asking drivers show readiness to think about data. Record the tally of predictions to compare with at the end of the unit.
Observe Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners work in groups of four (roles: Achieng - recorder for the data sources, Kiprop - reader of the cards, Njeri - sketcher, and one spokesperson). Each group receives three printed or board-copied items. Item 1 is the officer's claim with the facts: 8,000 m, 21 crossing points, about 400 m average. Item 2 is the near-miss	Before groups start, say: 'You are detectives. Every piece of information on your table is a clue.' Circulate and ask each group: 'Which of these clues did someone measure, which did someone ask, and which did someone write down?' WAIT TIME 10-15 seconds after each question. Prompt groups that stall: 'If you wanted to	Evidence table (claim-evidence-source analysis) with a gallery-style share. Sorting real clues into survey, observation and records builds the idea that data comes from different sources, each with different strengths. Asking what a source cannot tell us develops critical thinking about the quality of evidence.	Look for: (1) groups correctly classify the near-miss log as records and counting pedestrians as observation; (2) groups give a Thika Road example for each source that is realistic and safe, e.g. asking learners and drivers (survey) rather than crossing the road; (3) groups notice that the near-miss log does not say where

 EXERCISE: Calculating the mean Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	log kept for one week by matatu drivers led by Mzee Otieno: 12 near-misses were recorded; the drivers say most happened in one stretch around the school gate (the full breakdown of where they happened is withheld until a later lesson). Item 3 is a set of three Source Cards: Survey (asking people questions, e.g. asking 30 learners how they get to school and where they cross), Observation (watching and counting, e.g. standing safely at a footpath and tallying pedestrians crossing away from zebra crossings and measuring distance between crossing points with a trundle wheel or tape measure), and Records (information already written down, e.g. the matatu drivers' near-miss log, county road records, police accident records, school registers). Groups complete a table with three columns: 'Source', 'Thika Road example', 'What it could tell us about the officer's claim'. They also write what they notice about the near-miss log: what kind of information is it, who collected it, and what is missing. Groups also note a puzzle: 'The officer says well served, the driver says danger in one stretch. How can both be told by data?'	check whether the road is well served, would the officer's 400 m tell you where the gaps are long?' (do not answer it). Ask: 'What does the near-miss log tell us? What does it not tell us?' After 10 minutes, cold-call 3 groups to share one row of their table, e.g. 'Group 2, give us a survey example on Thika Road.' Use the discussion to clarify: a source gives data, and data can be gathered by asking, watching or looking at records. Note that the near-miss log has 12 entries, and ask: 'Is 12 a lot? What would you compare it with?' Accept all ideas and record them on the board under 'Observe: what we notice'. Introduce the term undesignated crossing: 'A pedestrian crossing the road away from a zebra crossing or footbridge.' Do not give pedestrian counts or gap lengths.		on the road each event happened or how many pedestrians cross; (4) at least one learner per group voices that the average 400 m does not show whether distances are equal. Use a checklist with the four indicators. If a group cannot classify the sources, ask them to say how each piece of information was obtained (asked, watched, or read).
Explain Phase  VIDEO: Range and mid-range <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Pairs build a working meaning of data and statistics. Step 1 (5 minutes): each pair writes in its own words: 'Data are ...' and 'Statistics is ...' using examples from Thika Road (e.g. the 8,000 m length, 21 crossing points, 12 near-misses, what learners see at the gate). Step 2: pairs take the four question prompts: 'How would you find out how far apart the	Say: 'In your own words, what are data? Write first, then talk.' Give 2 minutes silent writing, then pairs share. Cold-call 4 pairs, e.g. 'Kiprop and partner, give us your definition of data.' Use their words to build the class definition and do not simply dictate it. Ask: 'Is the officer's 400 m data? Where did it come from?' WAIT TIME 10-15 seconds. Press for the use of 'data'	Structured pair talk, then whole-class definition building from learners' own words (consensus discourse). Learners connect the abstract terms data and statistics to a concrete decision about their own road. Choosing a source and naming its weakness builds early understanding of data quality and sampling.	Look for: (1) definitions that include collecting facts or figures and using them to answer a question or make a decision; (2) correct use of 'data' as plural by at least half of the pairs; (3) a sensible match of source to question (observation for counting pedestrians and measuring gaps, records for near-misses, survey for opinions or habits); (4) mention of

 EXERCISE: Effects of shifting, adding, & removing a data point <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	crossing points really are?', 'How would you find out how many pedestrians cross away from the zebra crossings, and where?', 'How would you find out whether near-misses cluster in one stretch?', 'Who might disagree with our data, and why?' For each, they choose the best source (survey, observation or records), say why, and say what could go wrong (e.g. counting on only one morning, asking too few people, a log that drivers kept only for one week). Step 3: the class agrees a shared definition on the board: Data are facts, figures or observations collected about something; statistics is the collecting, organising, representing and interpreting of data to make decisions. Learners also state that the officer's 400 m is itself a figure that came from data (the length of the road and the number of crossing points).	as plural: 'These data show...' Ask the follow-up: 'If we only knew the officer's figure, what else would we need to know?' When pairs choose sources, challenge them: 'You chose observation. Where will you stand? How will you keep safe? On which day and time?' Highlight integrity: 'How do we make sure our tallies are honest and accurate?' Stress that statistics is used in day-to-day life for decisions, such as a county deciding where to build a footbridge, a farmer in Kericho deciding when to pick tea, or a matatu sacco choosing routes. Name the Sub-strand key idea: 'To decide whether the road is well served we must collect data, organise it, represent it and interpret it.' Do not teach calculation of mean, median or mode; if asked, say 'We will test this in Lesson 3'.		at least one limitation such as a single morning or a small sample; (5) learners recognising that the officer's average is itself data-based. Use a thumbs up, sideways, down check on 'I can explain what data are and name three sources'. Pairs who show sideways or down get a follow-up question on a Thika Road example.
Driving Question Board (DQB) Creation  VIDEO: Creating a bar graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Graph data on line plots <i>Related topic: not an exact match for this lesson.</i>	Learners build the Driving Question Board. The driving question is placed at the top: 'How can we use data from our own road to show where pedestrians are really at risk, and why does one average not tell the whole story?' Three columns below carry the KICD key inquiry questions: 'What is statistics?', 'How do we represent data?', 'How do we use statistics in day-to-day life?' Each learner writes 2 questions on sticky notes (one per note) arising from the phenomenon, e.g. 'Why do pedestrians cross away from the zebra crossing?', 'Are the crossing points evenly spaced?', 'Where exactly do the near-misses happen?', 'Can one number describe a whole road?', 'How do	Say: 'A good question is one that data can help us answer.' Model two examples on the board, one strong ('Where along the 8,000 m are the gaps between crossing points longest?') and one that needs rewording ('Is the officer bad?'), and ask learners to say what makes the difference. Give 3 minutes silent writing time. Invite 5 learners to post one note and read it aloud; cold-call a further 3 to say which column it belongs to and why. WAIT TIME 10-15 seconds when asking where a question belongs. Reword questions with the learner, not for them. Point out that many notes about 'how to show it in a picture' belong under 'How do we represent data?'. Place the board where it stays	Driving Question Board with question clustering. Learners' own questions become the learning agenda, which gives them ownership and a visible record of growing understanding. Sorting into the three KICD key inquiry questions links their curiosity to the sub-strand outcomes.	Look for: (1) the proportion of notes that are testable by data rather than opinions; (2) notes that show curiosity about spread or location (e.g. 'Are all gaps the same length?') which signal readiness for the main idea; (3) correct placement of notes under the three KICD questions; (4) learners who give a reason for a placement. Record the number of notes per column and keep a photograph or copy of the board. Notes that need rewording can be coached individually while learners work.

Source: Khan Academy (English - US curriculum) Search ARES for similar readings	we show data in a picture?' Learners post their notes under the column they fit best, or in a 'Not sure yet' area. In groups they cluster similar notes and give each cluster a heading. The class chooses 3 questions to investigate first and marks them with a star.	visible during the unit. Say: 'We will come back to this board in every lesson and move notes when we can answer them.' Use the learners' questions to connect forward: counting and tallying (Lesson 2), which average to trust (Lesson 3), and graphs (Lessons 6 and 7). Do not promise answers yet.		
Model Building Phase  VIDEO: Reading a line plot with fractions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Scientific method and data analysis <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Individually, learners draw an Initial Model on a sheet of paper or in the exercise book. They draw a line to represent the 8,000 m stretch of Thika Road from Roysambu stage towards Ruiru, mark the school gate at the 4,000 m mark, and place 21 small marks for crossing points in the way they think they are spaced (they may spread them evenly or not). They add where they think pedestrians cross away from the crossings and where near-misses happen, using arrows or shading. Beneath the drawing they write: (1) 'The officer's average of 400 m is ...' (true / false / true but not enough, with a reason); (2) 'I think the data would show ...'; (3) 'Three things I want to find out.' Pairs compare models and mark one similarity and one difference. A few volunteers display their models on the wall next to the DQB. Learners keep the model in their books for revision in later lessons.	Say: 'Draw what you think the road really looks like. There is no wrong model today, because this is a first idea we will test.' Remind learners the 400 m average is true as arithmetic: 8,000 m shared among the gaps does give about 400 m. Say: 'Her figure is true but may be incomplete. Show in your model whether the crossing points are evenly spaced or not.' Do not correct models. Circulate and ask: 'Why did you put the crossings there?' WAIT TIME 10-15 seconds. Cold-call 3 learners to describe their models in one sentence. Ask the class: 'How many models show all crossings evenly spaced? How many show clusters or long empty stretches?' Count hands and write numbers on the board. Close by recalling the three data sources and say: 'Next lesson we start collecting and organising data from Thika Road using tally marks. Think about how we will keep accurate records.' Collect the Predict Cards and models for later comparison. Give a short take-home task: ask one adult (a parent, guardian or matatu driver, if safe and possible) where on Thika Road people cross away from the crossings, and record the answer, as a first small survey.	Initial model drawing with pair comparison. Putting ideas into a spatial model forces learners to make their thinking about spacing and risk explicit, so that it can be revised against evidence as the unit proceeds. Comparing models reveals different assumptions to be tested.	Look for: (1) models that show uneven spacing of crossing points or clusters of pedestrians, showing an early grasp that an average hides variation; (2) models with evenly spaced crossings and a claim that the road is safe, indicating the misconception to address in Lessons 3 to 8; (3) a 'true but not enough' judgement with a reason; (4) 'things I want to find out' that refer to counts, distances or locations. Collect models and sort into three piles (even, clustered, unsure) to inform Lesson 2 grouping. Use a quick exit question: 'Name one source of data and one thing it could tell us about Thika Road.'

D. TEACHER REFLECTION

After the lesson, review the Predict Cards and initial models. How many learners predicted 'safe', 'not safe' and 'safe in some places'? Did learners treat Ms Abdi's claim respectfully as true but possibly incomplete, or did some conclude she was wrong or careless? Which groups could match a data source to a question, and which confused survey, observation and records? Did learners use 'data' as a plural? Did the DQB notes include testable questions, and did any under 'How do we represent data?' show readiness for graphs later? Which learners were quiet and need a role in Lesson 2 (for example Achieng's tally group)? Are safe survey arrangements with parent or guardian volunteers confirmed for the data collection? Note any learner who was affected by road accidents and offer support. Adjust Lesson 2 groupings and the tally sheet design in light of this evidence.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed the County Road Safety Officer's claim that the average distance between crossing points on the 8,000 m stretch of Thika Road is about 400 m (21 crossing points), and that this is true. We also saw learners crossing away from the crossings at our school gate, and we read the log of 12 near-misses kept by Mzee Otieno and other matatu drivers, who say most happen in one stretch.
What did I learn?	We learned that data are facts, figures or observations collected about something, and that statistics is collecting, organising, representing and interpreting data to make decisions. We learned three sources of data: surveys (asking), observation (watching, counting and measuring) and records (information already written down). We learned that an undesignated crossing is a pedestrian crossing away from a zebra crossing or footbridge.
How does this explain the phenomenon?	We can explain that the officer's average of 400 m may be true but incomplete, because one number may not show whether the crossing points are evenly spaced or where pedestrians are at risk. To test the claim we need to collect our own data on the gaps between crossing points and the pedestrians crossing away from them, and we will keep adding to the Driving Question Board and our model as we learn.

LESSON 2 (40 minutes (one lesson; the Thika Road morning data is supplied as the class's recorded tally sheets)): Counting the Road: Tallies and Frequency Tables

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners collect data with tally marks and organise ungrouped data into frequency distribution tables, so that the 20 gaps and the pedestrian counts on the 8,000 m stretch of Thika Road can be examined beyond the officer's single average of 400 m.
Knowledge	By the end of the lesson the learner should be able to: (a) collect data from real-life sources using tally marks; (b) draw a frequency distribution table for ungrouped data (gap lengths and pedestrians per gap); (c) check that total frequency equals the number of observations (20 gaps) and that the totals match the whole road (8,000 m; 182 pedestrians).
Skills	Recording with tallies in bundles of five; sorting and ordering raw data; building frequency tables with columns for value, tally and frequency; checking totals; reading a frequency table to spot unusual values. NGSS-style practices embedded: Planning and Carrying Out Investigations, Analyzing and Interpreting Data, Using Mathematics and Computational Thinking, Constructing Explanations.
Attitudes	Integrity: the learner collects and records data accurately. Unity: the learner works in groups to tally and tabulate. Core competencies: Critical thinking and problem solving; Communication and collaboration. PCI: Road safety; Life Skills (Decision Making).
Key Inquiry Question	What is statistics? How do we represent data? How do we use statistics in day-to-day life? (Focus today: How do we represent data?)
Purpose in Storyline	Lesson 1 raised the officer's claim (about 400 m average; 8,000 m; 21 crossing points), the near-miss log, and the plan to measure gaps and count pedestrians. Lesson 2 turns that plan into organised evidence: two ungrouped frequency tables for the 20 gaps and the pedestrian counts. These tables are the input for Lesson 3, where mean, median and mode are calculated. No averages are calculated in this lesson, only organised.
Safety Notes	All data is collected from records and from the classroom tally sheets; no learner goes onto Thika Road during the lesson. For any real field counting, use a safe spot away from the carriageway with a teacher or adult, never step on the road, and wear bright clothing. Handle tape measures and trundle wheels carefully.

B. LESSON OVERVIEW

Ruaraka Hills Secondary School is on Thika Road, Nairobi. Mwalimu Wanjiru reminds learners of the plan from Lesson 1: to test Ms Halima Abdi's claim, the class must count the 21 designated crossing points, measure the 20 gaps between them in metres, and count pedestrians crossing at undesignated spots in each gap during the Tuesday 6:30-7:30 am count. First, learners practise tallying with a quick classroom activity (counting matatu stage names or ugali, githeri and mandazi choices from a mini-survey). Then group leaders Achieng (tallies), Kiprop (tables) and Njeri (checking and graph preparation later) lead groups in working from the raw gap list: G1 200 m, G2 150 m, G3 250 m and so on to G20 200 m. Groups build the ungrouped frequency table for gaps (100:3, 150:2, 200:4, 250:3, 300:2, 600:1, 700:1, 800:1, 1000:1, 1050:1, 1100:1; total 20) and for pedestrians per gap (1:2, 2:3, 3:4, 4:3, 5:2, 15:1, 18:1, 22:1, 26:1, 28:1, 31:1; total 20 gaps). They check that gaps sum to 8,000 m and pedestrians to 182 using the frequency table, then notice that most values are small but a handful are very large. The officer's claim is true but incomplete; today learners see the first hint of why, without computing any average.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Frequency tables & dot plots Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Frequency Tables to Organize and Display Data (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the plan from Lesson 1. Mwalimu Wanjiru shows the raw list of the 20 gap lengths in order along the road (G1 to G20) on the board, unsorted and unorganised. Individually (2 minutes) they write a prediction in their books: 'If I organise these 20 gaps into a table, what will I see? Will most gaps be near 400 m, or not?' They also predict: 'Which gap lengths will appear most often?' Then they share with a partner and one group sketches the table they expect.	Say: 'Last lesson we planned to measure gaps and count pedestrians. Here is what the survey team recorded. Before we organise it, what do you expect to find? Write one prediction and one reason.' Display the raw list without commentary. Give WAIT TIME of 10-15 seconds after asking: 'Does a list in this order make it easy to say which gaps are common?' Cold-call 4 learners (one from each corner of the room) to read their predictions without evaluating them. Ask: 'Who predicted that most gaps would be near 400 m? Raise your hand.' Count and record on the board. Say: 'Keep your prediction; we will test it with a table.'	Predict-then-test with a Think-Pair-Share: Writing a prediction first commits learners to an idea about the officer's average, so the later table gives them something to confirm or challenge. It builds understanding because they feel the need for an organising tool when the raw list looks messy.	Observable: (1) Every learner has a written prediction with a reason before discussion. (2) At least 4 distinct predictions are shared. (3) Note the number of learners who expect most gaps near 400 m (this is revisited in the Explain phase). (4) Listen for learners who say a list is 'hard to read' or 'too messy', showing readiness for tallies.
Observe Phase  VIDEO: Frequency tables & dot plots Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Frequency Tables to Organize and Display Data Practice Source: CK-12  Search ARES for similar readings	Part A (8 minutes): Tally practice. Groups of 5 conduct a mini-survey of their own: 'Which food did you eat for lunch yesterday: ugali with sukuma wiki, githeri, rice and beans, or mandazi?' Achieng's role in each group records tally marks in bundles of five (/). Groups then write the frequency for each food. Part B (14 minutes): Groups work with the Thika Road tally sheets. Using the raw list of 20 gap lengths (G1 200 m, G2 150, G3 250, G4 100, G5 300, G6 200, G7 100, G8 250, G9 700, G10 1000, G11 1100, G12 800, G13 1050, G14 600, G15 200, G16 150, G17 100, G18 250, G19 300, G20 200), they make a table with columns: Gap length (m) Tally Frequency. They list each different value in order from smallest to largest, tick off each of the 20 gaps once as they tally, and total the frequency column. They repeat for	Circulate with a checklist. Say: 'Tally in fives: four vertical strokes and a diagonal line across. Why do we bundle in fives?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Use these prompts: 'Show me how you make sure no gap is counted twice.' (e.g., strike through each value on the raw list as you tally). 'What should the frequency column add up to? Why?' (20, one for each gap). For the 21 designated crossing points, ask: 'Why are there 20 gaps and not 21?' (Between 21 points there are 20 spaces.) If a group's total is not 20, say: 'Your frequencies add up to 19. Which gap might you have missed? Go back to the list and check G1 to G20 in order.' Do not give the answer. Once groups finish the gap table, pause the room and ask: 'Add up gap length x frequency in your head roughly. Does it come near 8,000 m?' (Full	Data organising routine (raw list to tally to frequency table) with a Check-the-Total protocol: moving from messy data to a structured table shows how organisation reveals patterns. The total check (20 gaps, 8,000 m, 182 pedestrians) models accuracy and Integrity, so groups trust their table before drawing conclusions.	Observable indicators: (1) Tally marks are grouped in fives and counted correctly. (2) Values are listed in ascending order with no repeats. (3) The gap table matches 100:3, 150:2, 200:4, 250:3, 300:2, 600:1, 700:1, 800:1, 1000:1, 1050:1, 1100:1 with total frequency 20. (4) The pedestrian table matches 1:2, 2:3, 3:4, 4:3, 5:2, 15:1, 18:1, 22:1, 26:1, 28:1, 31:1 with total 20 gaps. (5) Groups verify that gaps add to 8,000 m and pedestrians to 182. Use a checklist to record groups who need support (e.g., missed values, wrong order).

	pedestrians per gap (G1 3, G2 2, G3 4, G4 1, G5 5, G6 3, G7 2, G8 4, G9 18, G10 26, G11 31, G12 22, G13 28, G14 15, G15 4, G16 2, G17 1, G18 3, G19 5, G20 3). Kiprop checks the table is ordered and Njeri checks the frequency total is 20.	calculation of mean is Lesson 3; here only check the total.) Pose the totals check: 'Multiply each gap length by its frequency and add; what total do you expect, and why?'		
Explain Phase  VIDEO: Ungrouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Creating frequency tables Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups compare their frequency tables with a neighbouring group, then with the class master table on the board. They look at the gap table and answer in writing: (1) Which gap length appears most often? (200 m, 4 times.) (2) How many gaps are 300 m or shorter? (14 of the 20.) (3) How many gaps are longer than 300 m, and how long are they? (6 gaps, all from 600 m to 1,100 m.) They then look at the pedestrian table: the counts of 1 to 5 appear for 14 gaps, but 15, 18, 22, 26, 28 and 31 each appear once. Learners discuss: 'Does the table support the officer's statement that the road is well served? What does the table show that one number, 400 m, hides?' They link the large gaps and large pedestrian counts: the six large counts match the six long gaps G9 to G14, though they do not yet compute any average. They write a claim-evidence-reasoning statement.	Say: 'Look at your table, not at the raw list. What can you see in 30 seconds that you could not see in the list?' WAIT TIME 10-15 seconds. Cold-call 5 learners with different observations. Press for evidence: 'Which row of the table tells you that?' Then ask: 'Ms Halima Abdi said the average gap is about 400 m. Look at your frequency column: how many gaps are actually close to 400 m?' (None are exactly 400 m; 14 are 300 m or shorter and 6 are 600 m or more.) Say: 'The officer's claim is true but incomplete. The 400 m is correct arithmetic (8,000 m / 20 gaps) but the table shows two groups of gaps.' Ask: 'Mzee Otieno says most near-misses happen in one stretch. Which rows of the pedestrian table might that be?' Hold back Lesson 3 content: if a learner offers 'mean, median, mode', say: 'Good words. Tomorrow we calculate them; today we ask what the table shows us.' Collect 3 CER statements aloud.	Claim-Evidence-Reasoning with a Notice-and-Wonder routine: learners notice features of the tables (clusters, gaps in values, repeated values, outliers) and then connect them to the phenomenon. The strategy builds understanding because the evidence comes directly from their own organised data, which makes the officer's claim and the drivers' remarks testable.	Observable: (1) Learners cite frequencies as evidence (for example, 'the 200 m gap appears 4 times' or '14 gaps are 300 m or shorter'). (2) CER statements mention both the correct arithmetic of the 400 m and the pattern in the table. (3) Learners distinguish the six long gaps and six high pedestrian counts from the rest. (4) Listen for the misconception that 'the table proves the road is dangerous everywhere' or 'the officer is wrong'; correct with 'true but incomplete'. Exit prompt on a slip: 'Write one thing the frequency table shows that a single average does not.'
Driving Question Board (DQB) Creation  VIDEO: Statistics intro: Mean, median, & mode <i>Related topic: not</i>	Learners return to the DQB started in Lesson 1, with the three KICD key inquiry questions: What is statistics? How do we represent data? How do we use statistics in day-to-day life? Each group adds sticky-note evidence under 'How do we represent data?' (for example, 'Tally marks and	Say: 'Look at our Driving Question: How can we use data from our own road to show where pedestrians are really at risk, and why does one average not tell the whole story? What did today's tables add?' WAIT TIME 10-15 seconds. Cold-call 3 groups to read one sticky note each and say	Driving Question Board with question clustering: sorting questions makes the learning pathway visible. It builds understanding by showing learners that today's tables raise new questions that later tools (averages, grouping, graphs) will answer.	Observable: (1) Each group posts at least one evidence note and one new question. (2) Questions refer to the data (gaps, pedestrians, near-misses) rather than generic topics. (3) Learners correctly link notes to the key inquiry questions. (4) Notice whether at least two groups ask

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Reading dot plots & frequency tables Source: Khan Academy (English - US curriculum) Search ARES for similar readings	frequency tables organise 20 gaps and 182 pedestrians') and writes at least one new question on a sticky note, such as 'Which gap length is typical?', 'How can we find the middle value?', 'The table has many values; can we group them?' or 'Where on the road do the large pedestrian counts happen?' They place the notes on the board and sort them into 'Can answer soon' and 'Need more tools'.	which key inquiry question it connects to. Say: 'Who can link this note to Lesson 3, where we will calculate mean, median and mode?' Highlight any question about averages and mark it for next lesson. Cluster questions about grouping into classes and tag them for Lesson 4. Model a good question: 'A strong question is one the data could help us answer.'		about averages or about where the risk is greatest.
Model Building Phase VIDEO: Histograms <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Frequency Tables to Organize and Display Data (Flexbook) Source: CK-12 Search ARES for similar readings	Learners revise their initial model from Lesson 1 in their books. The model is a simple diagram of Thika Road from the Roysambu stage towards Ruiru with 21 crossing points, now annotated with data-handling steps: Collect (tally) then Organise (frequency table) then Check (totals 20 gaps, 8,000 m, 182 pedestrians). They add a mini frequency table of gap lengths to the model and mark the six long gaps (600 m to 1,100 m) beside the pedestrian counts of 15 to 31. They write a short note: 'Our table shows that the officer's 400 m is correct but many gaps are short and a few are very long.' They add one thing they still cannot explain, for example 'which number best represents the typical gap'.	Say: 'Revise your model. What new piece of evidence goes into it today, and what question does your model still not answer?' WAIT TIME 10-15 seconds. Circulate and ask 3 learners: 'Where in your model would a reader see that the road is not evenly served?' Push for specifics: 'Point to the part of your table that shows it.' Ask two volunteers to share their revised models. Close by saying: 'Next lesson we ask which measure, mean, median or mode, tells the truth about these 20 gaps. Keep your tables safe; you will use them.' Remind learners to keep tally sheets and tables in their project folders.	Cumulative model revision (gotta-have-it checklist): adding the data-handling process and the table to the evolving model shows how evidence changes the explanation. It builds understanding because learners see the model becoming more precise and recognise its remaining gaps.	Observable: (1) Revised models include the frequency table or its key rows. (2) Models show the 20 gaps and mention the 8,000 m and 182 pedestrians totals. (3) The model distinguishes short gaps from the six long ones. (4) Learners identify a remaining question (for example, which average to use). Record who can state, in their own words, why a tally and frequency table help in testing the officer's claim.

D. TEACHER REFLECTION

After the lesson, reflect: Did every group's tables total 20 gaps, 8,000 m and 182 pedestrians, and which groups needed help with ordering values or missed a gap? Which learners confused frequency (how many gaps) with the gap value (metres)? Did learners say 'the officer is wrong', and how did I reframe it as 'true but incomplete'? Which DQB questions about averages or grouping should open Lesson 3 and Lesson 4? What would I change about the tally practice to give quieter learners more speaking time? Remember this is a one-morning sample (Tuesday, 6:30-7:30 am) and may need repeating.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We tallied the 20 gaps between the 21 designated crossing points on the 8,000 m stretch of Thika Road and the pedestrians crossing at undesignated spots in each gap (182 in total). Our frequency tables showed that gap lengths of 100 m to 300 m are common (14 of the 20 gaps), while six gaps are between 600 m and 1,100 m, and six gaps had pedestrian counts of 15 to 31.
What did I learn?	Tally marks (in bundles of five) help us collect data, and a frequency distribution table for ungrouped data organises it so that we can check the total (20 gaps) and see which values are common or unusual.
How does this explain the phenomenon?	The officer's 400 m average is true but incomplete. Our frequency table shows that most gaps are short while a few are very long, which suggests that a single number hides where pedestrians may be at risk. We need more tools, such as mean, median and mode, to describe the data fairly.

LESSON 3 (80 minutes (2 x 40-minute periods)): Mean, mode and median: which one tells the truth?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners calculate the mean, mode and median of the ungrouped gap data and pedestrian counts from Thika Road (20-gap sample) and compare the three measures to judge whether the officer's claim of a 400 m average fully represents the road.
Knowledge	By the end of the lesson the learner should be able to: (a) determine the mean, mode and median of ungrouped data (KICD Specific Learning Outcome 3.1 c); (b) use a frequency distribution table to calculate the mean as the sum of ($f \times x$) divided by the sum of f ; (c) state the range as the highest value minus the lowest value; (d) explain why the mean of 400 m is correct but incomplete, since the median is 250 m and the mode is 200 m; (e) judge which measure best represents the data.
Skills	Calculating with frequency tables (mean = sum of $f \times x$ / sum of f); ordering data and locating the middle value (mean of the 10th and 11th values for $n = 20$); NGSS/SEP-style practices: analysing and interpreting data, using mathematics and computational thinking, constructing explanations from evidence, engaging in argument from evidence. Core competencies: Critical thinking and problem solving; Communication and collaboration.
Attitudes	Integrity: the learner calculates and records values accurately and reports them honestly. Unity: the learner works in groups and shares tasks (tallies, tables, calculations). Respect for the officer: the claim is true but incomplete, never 'wrong'. PCI: Road safety; Life Skills (Decision Making).
Key Inquiry Question	KICD key inquiry questions: 1. What is statistics? 2. How do we represent data? 3. How do we use statistics in day-to-day life? Lesson focus: How do we use statistics in day-to-day life? Can one average tell us where pedestrians on Thika Road are at risk?
Purpose in Storyline	Lesson 2 organised the 20 gaps and the pedestrian counts into ungrouped frequency tables (totals 8,000 m and 182 pedestrians). This lesson uses those tables as the input for calculating the three measures. Learners confirm that the officer's mean of 400 m is arithmetically true, then discover that the median (250 m) and the mode (200 m) tell a different story, and the range (1,000 m) shows a wide spread. This creates the need for Lesson 4, where the gaps are grouped into classes because the range is so large.
Safety Notes	This is a classroom data lesson using data already collected, with no learners on the road. Remind learners that any future data collection on Thika Road must be done with the teacher, in a group, away from traffic, wearing bright clothing, and only from the pavement or a footbridge. Do not name individual pedestrians or record faces. Calculators must be shared fairly.

B. LESSON OVERVIEW

Ruaraka Hills Secondary School, on Thika Road in Nairobi, is testing Ms Halima Abdi's statement that 'the average distance between crossing points is about 400 m, so the road is well served.' Using the Lesson 2 frequency table of the 20 gaps (100:3, 150:2, 200:4, 250:3, 300:2, 600:1, 700:1, 800:1, 1000:1, 1050:1, 1100:1), learners first predict whether the mean, median and mode will be the same. Under Mwalimu Wanjiru's guidance, group leaders Achieng (checking tallies), Kiprop (calculations) and Njeri (recording and display) help groups calculate the mean (400 m), median (250 m), mode (200 m) and range (1,000 m). They repeat the process for the pedestrian counts per gap (mean 9.1, median 4, mode 3, range 30). In the Explain phase, learners argue from evidence about which measure best represents the typical gap, concluding that the officer's mean is true but incomplete because a few very long gaps (600 m to 1,100 m) pull it above the median. They add the three measures to the DQB and revise their initial model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the officer's claim and the Lesson 2 frequency table of 20 gaps shown on the board. Individually, then in pairs, they answer on a prediction card: 'Will the mean, median and mode of the 20 gaps all be about 400 m? Which do you think will be biggest? Smallest?' They write one sentence of reasoning and a number estimate for each measure. They also quickly use the ugali-and-sukuma-wiki analogy: five households in Kisumu pay KSh 20, 20, 30, 30 and 400 for a week's vegetables; what is the 'typical' payment? Groups then post their predictions on the class chart.	Display the frequency table from Lesson 2 and read the officer's claim aloud. Say: 'Ms Halima Abdi says the average gap is about 400 m. Before we calculate anything, what do YOU think the mean, median and mode will be? Write your prediction first, then your reason.' Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds after asking: 'Do you think they will all be the same?' Cold-call 4 learners from different groups (not volunteers) to share and ask a fifth learner to restate one prediction in their own words. Pose the vegetable-price analogy and ask: 'Which number would you tell a new neighbour is the normal price?' Record all predictions on the chart without judging them. Say: 'The officer is not careless. Let us find out together what the data say.'	Predict-and-Justify (Prediction Chart): committing to a written prediction with a reason exposes learners' ideas that 'average' always means 'mean' and that all averages agree. Returning to the chart after calculation creates cognitive conflict that motivates the comparison of measures.	Look for: (1) at least 80% of learners have written a number and a reason for each measure; (2) note how many predict all three equal or near 400 m (misconception: all averages are the same); (3) note learners who say the mean could be pulled by very large values (strong prior idea); (4) listen in the vegetable-price example for learners who say 'KSh 30' is more typical than the mean. Record counts on the teacher checklist for use in the Explain phase.
Observe Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Ungrouped Data to Find the Mean (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of four (Achieng checks the table against the tally, Kiprop leads calculations, Njeri records and draws the display, one learner is the reporter) work with the Lesson 2 ungrouped frequency tables. Task 1: Find the total of the f x x column: $100 \times 3 = 300$, $150 \times 2 = 300$, $200 \times 4 = 800$, $250 \times 3 = 750$, $300 \times 2 = 600$, 600, 700, 800, 1000, 1050, 1100, so the sum is 8,000 m, and divide by the total frequency of 20 to get a mean of 400 m, which confirms the officer's arithmetic. Task 2: Identify the mode, the value with the highest frequency, 200 m (it appears 4 times). Task 3: Write the sorted list (100,100,100,150,150,200,200,20	Circulate with a checklist. Use prompts such as: 'Show me where the 8,000 m comes from in your table.' 'Why do we divide by 20 and not by 11, the number of different values?' 'How many values are below your median? How many above?' Give WAIT TIME of 10-15 seconds after each question before offering hints. Ask groups that finish early: 'Where does the school gate (4,000 m mark) sit relative to the six long gaps?' Pause the class at the 10-minute mark and cold-call 3 groups to report their mean; if any report a different mean, ask the whole class to find the error ('Check the f x x column: did you	Structured Data Analysis with a Class Results Table: learners compute the same three measures from the same frequency table, so differences between measures come from the data and not from errors. The class results table (gap data and pedestrian data side by side) makes patterns visible: for both data sets the mean is larger than the median and the mode.	Observable indicators: (1) learners correctly build the f x x column and obtain 8,000 m and 182; (2) they divide by total frequency 20, not by the number of classes; (3) they order the data before finding the median and use the mean of the 10th and 11th values; (4) they identify the mode as the value (200 m, or 3 pedestrians), not the frequency (4); (5) correct range of 1,000 m. Use a quick tally on the checklist; groups with two or more errors get a mini-conference at the next stage. Collect Njeri's recording sheets to check results.

	0,200,250,250,250,300,300,600,700,800,1000,1050,1100), count to the 10th and 11th values (both 250 m), and calculate the median as 250 m. Task 4: Find the range, 1,100 - 100 = 1,000 m. Task 5: Repeat for pedestrian counts per gap (1:2, 2:3, 3:4, 4:3, 5:2, 15:1, 18:1, 22:1, 26:1, 28:1, 31:1): the sum of $f \times x$ is 182, so the mean is $182/20 = 9.1$ pedestrians per gap; the mode is 3; the sorted 10th and 11th values are both 4, so the median is 4; the range is $31 - 1 = 30$. Groups also count how many of the 20 gaps are 300 m or shorter (14) and how many are over 500 m (6).	multiply frequency by value?'). Stress the mean equals $8,000/20 = 400$ m, matching the officer. Hold up the sorted list and ask learners to 'count in from both ends' until they meet in the middle. Have Kiprop's group demonstrate on the board while Njeri's group records the three measures on a class results table. Ensure calculators are shared and that learners also check mentally: the mean of the pedestrian counts should land between the smallest and largest value.		
Explain Phase  VIDEO: Mean, median, & mode example - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Ungrouped Data to Find the Mean Practice Source: CK-12  Search ARES for similar readings	Whole-class discussion using the results table: gap mean 400 m, median 250 m, mode 200 m, range 1,000 m; pedestrians mean 9.1, median 4, mode 3, range 30. Learners revisit their predictions and then take part in an 'Argument from Evidence' task: 'Which measure best represents a typical gap on Thika Road? Which would the officer, a matatu driver like Mzee Otieno, and the school principal each prefer to quote?' Each group writes a claim, evidence (numbers from the table, e.g. 14 of the 20 gaps are 300 m or shorter, only 6 gaps exceed 300 m and all six are over 500 m), and reasoning (the six very long gaps of 600 m to 1,100 m pull the mean up to 400 m, a value that only 6 of 20 gaps exceed). Learners explain in their own words why the mean of 400 m is true but incomplete, since a mean of 400 m describes the total spread over the road and not any actual typical gap: no gap lies between 300 m and 600 m. They compare with the pedestrian	Return to the prediction chart: 'Which predictions were close? What surprised you?' Ask: 'The officer said 400 m. You also found 250 m and 200 m. Who is right?' WAIT TIME of 10-15 seconds, then use Think-Pair-Share, then cold-call 4 groups, each adding one new piece of evidence without repeating. Press for precision: 'How many gaps are actually near 400 m?' (none between 300 m and 600 m). 'If we replaced the six longest gaps with 250 m each, what would the mean be?' (letting learners estimate that it falls sharply). Model the sentence frame: 'The officer's claim is true but incomplete because ...' Never say 'wrong'. Ask: 'Which measure would you give the county roads office if you wanted them to build more crossings? Which one would hide the problem?' Connect to the phenomenon: 'What does the mean not tell us about WHERE the gaps are long?' Lead learners to see that the three measures describe the centre of the data but	Argument from Evidence (Claim-Evidence-Reasoning) combined with Compare-and-Contrast: learners use the three measures as evidence, so their conclusion about 'the typical gap' is justified by numbers rather than opinion. Comparing the gap and pedestrian results shows that the mean is sensitive to extreme values while the median and mode are not, building the key idea that no single average tells the whole story.	Look for: (1) claims supported with at least two numbers from the table; (2) the explanation that the six long gaps pull the mean above the median and mode; (3) correct statement that the officer's mean is arithmetically true; (4) use of respectful language ('true but incomplete'); (5) the idea that the mean hides where the long gaps are. Use a 1-3 rubric on the CER sheet (1 = claim only, 2 = claim with one piece of evidence, 3 = claim, evidence and reasoning). Exit ticket question: 'Why is the median a better description of a typical gap than the mean for this road?' Aim for at least 75% scoring 2 or higher.

	counts: mean 9.1 against median 4 and mode 3, showing that a handful of gaps carry most of the pedestrians. They also write a short statement of when the mean, median or mode is most useful (mean for totals such as 8,000 m, median when a few extreme values distort the picture, mode for the most common case).	do not show location along the road, setting up the need for grouping and graphs. Ask 2 learners with different positions to respond to each other's reasoning ('I agree/disagree with ... because ...').		
Driving Question Board (DQB) Creation  VIDEO: Mean and standard deviation versus median and IQR <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the Driving Question Board started in Lesson 1. Under the three KICD key inquiry questions (What is statistics? How do we represent data? How do we use statistics in day-to-day life?), each learner writes at least one sticky note: either an answer found today (for example 'The mean of the 20 gaps is 400 m, the median is 250 m and the mode is 200 m') or a new question (for example 'Where along the road are the long gaps? Do they match where the near-misses happen?' 'Is the median also 250 m if we group the gaps into classes?' 'How can we show the range of 1,000 m in a picture?'). Groups sort the notes into 'answered' and 'still to investigate', and link the new questions to upcoming lessons.	Say: 'Take two minutes of silence to write one thing you can now answer and one question that you still have.' WAIT TIME of 10-15 seconds before inviting sharing. Cold-call 3 learners to read their notes and place them on the DQB under the matching key inquiry question. Ask: 'Which of these questions needs a picture, not a number?' to point towards grouping (Lesson 4), and later graphs. Highlight any question about the location of the long gaps and the school gate at the 4,000 m mark. Move answered notes to the 'What we know' section and keep open questions visible for the next lessons.	DQB Tracking: seeing answered and open questions displayed publicly shows learners how their understanding is growing and gives them ownership of the next steps in the investigation.	Check the quality of questions: do they link to the phenomenon (Thika Road, near-misses, the long stretch)? Do at least 70% of learners add a note? Are answers stated with correct values and units (m, pedestrians per gap)? Note which key inquiry question receives the fewest notes and revisit it in Lesson 4.
Model Building Phase  VIDEO: Measures of spread: range, variance & standard deviation <i>Related topic: not an exact match for this lesson.</i>	Groups revise their class model that explains why an average can look reassuring while the road is risky. They add to the model: (1) the mean of 400 m, which is correct and comes from 8,000 m divided by 20 gaps; (2) the median of 250 m and mode of 200 m, which show most gaps are short; (3) the range of 1,000 m, which shows a few very long gaps; (4) the pedestrian mean of 9.1	Say: 'Add today's evidence to your model. Show how the mean, median and mode are connected to the road itself.' Provide sentence frames: 'The mean is 400 m because ...' 'The median is lower because ...' Give WAIT TIME of 10-15 seconds before groups begin drawing. Circulate and ask each group: 'What does your model say about the school gate at the 4,000 m mark?' (it falls inside	Cumulative Model Revision: updating a visible model after each lesson makes the link between evidence and explanation explicit, and lets learners see that today's evidence supports the officer's number but changes how much we trust it as a full description of the road.	Indicators: the model shows the mean, median and mode with correct values and units; includes an arrow or statement that long gaps pull the mean up; avoids saying the officer is wrong; contains at least one open question about location. Use a 3-point checklist: values correct, causal link shown, open question stated. Note groups that confuse 'mode' with the frequency for

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Ungrouped Data to Find the Mean (Flexbook) Source: CK-12 Search ARES for similar readings	compared with a median of 4 and mode of 3, which show that a few gaps carry many undesignedated crossers. Using arrows, they show that 'a few very long gaps pull the mean up' and mark a question: 'Where are the long gaps, and are the pedestrians there?' Groups present their revised model in 60 seconds to another group (gallery walk).	the long stretch, but today's evidence does not yet show this, so mark it as 'to be tested'). Cold-call 3 groups to explain one revision. Remind learners that this is a one-morning sample of 20 gaps on Thika Road (Tuesday, 6:30-7:30 am) and that it should be repeated. Close by saying: 'Next time we group the gaps into classes, because a range of 1,000 m is too wide to see the pattern in a single list.'	follow-up.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners calculate the mean from the frequency table correctly, especially the $f \times x$ column and dividing by 20? Which groups needed support? (2) How many learners could explain, in their own words, why the median (250 m) is lower than the mean (400 m)? (3) Did any learners treat the officer as wrong, and how did I redirect them to 'true but incomplete'? (4) Which examples (vegetable prices, Thika Road gaps) helped most? (5) What do I need to revisit in Lesson 4 so learners see why the range of 1,000 m calls for grouping? Record the names of learners who need extra practice finding the median of an even number of values.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We calculated the mean, mode and median of the 20 gaps between crossing points on an 8,000 m stretch of Thika Road (mean 400 m, median 250 m, mode 200 m, range 1,000 m) and of the pedestrian counts per gap (mean 9.1, median 4, mode 3, range 30). 14 of the 20 gaps are 300 m or shorter, and six gaps are 600 m to 1,100 m.
What did I learn?	Mean = sum of ($f \times x$) divided by sum of f ; median is the middle value of the ordered data (the mean of the 10th and 11th values for 20 values); mode is the most frequent value; range is highest minus lowest. The mean is affected by very large values, while the median and mode are not.
How does this explain the phenomenon?	The officer's claim of a 400 m average is true but incomplete: a few very long gaps pull the mean above the median (250 m) and mode (200 m), so one average hides the fact that most gaps are short and a few are very long. We still need to find where the long gaps are and how many pedestrians cross there.

LESSON 4 (40 minutes (one lesson; the table can be completed in pairs and finished as homework if needed)): Grouping the Gaps into Classes

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners group the 20 gap measurements from the 8,000 m stretch of Thika Road into classes of equal width, build a grouped frequency table and identify the modal class in pairs. This shows how grouping reveals the shape of the data that the single 400 m average hides.
Knowledge	By the end of the lesson the learner should be able to: (a) group collected data into appropriate classes and generate a frequency distribution table (KICD Specific Learning Outcome b: 'draw a frequency distribution table for grouped and ungrouped data'); (b) discuss in pairs and identify the modal class for grouped data; (c) explain why a range of 1,000 m (100 m to 1,100 m) makes an ungrouped table with 11 different values hard to read, and why a class width of 200 m is a sensible choice; (d) write classes as '200 to <400', with the lower boundary included and the upper boundary excluded.
Skills	Core competency: Critical thinking and problem solving (choosing class width and checking that every gap falls in exactly one class). Communication and collaboration (pairs agree on the modal class and justify it). NGSS-style practices: Analysing and interpreting data (spotting the pattern in a grouped table); Using mathematics and computational thinking (tallying and totalling frequencies); Constructing explanations (why the modal class does not locate the danger on its own).
Attitudes	Values: Integrity (record each gap in exactly one class and check the total is 20). Love (consideration for a partner when pairs discuss and identify the modal class). Unity (pairs share tasks fairly). PCIs: Road safety (the grouped table points to the stretch where children cross at undesignated spots); Life Skills (Decision Making) (choosing a class width is a decision that must be justified).
Key Inquiry Question	KICD key inquiry questions: 1. What is statistics? 2. How do we represent data? 3. How do we use statistics in day-to-day life? Lesson focus: How do we represent data? Which classes make the 20 gaps on Thika Road easy to read, and what does the modal class tell us and not tell us about where pedestrians are at risk?
Purpose in Storyline	Lesson 2 gave us the ungrouped frequency table (11 different gap values, total 20) and Lesson 3 showed a range of 1,000 m. With so many different values, the pattern is hard to see. Lesson 4 groups the 20 gaps into six classes of width 200 m (frequencies 5, 9, 0, 2, 1, 3) with modal class 200 to <400. This prepares for Lesson 5 (grouped mean and median) and Lesson 6 (histograms). It advances the driving question by showing in one glance that 14 of 20 gaps are under 400 m, while a separate cluster of long gaps sits at the right of the table. Do not calculate grouped mean or median here; those belong to Lesson 5.
Safety Notes	This lesson is classroom-based and uses the data already collected. If any learner revisits the road to re-measure a gap, they must go with Mwalimu Wanjiru in a supervised group, stay on the footpath behind the fence, never stand on the carriageway and never cross at undesignated spots. Handle tape measures and trundle wheels carefully. Sharing of data should respect the dignity of the community members and drivers, such as Mzee Otieno, who helped.

B. LESSON OVERVIEW

Mwalimu Wanjiru reminds the class of the officer's claim (Ms Halima Abdi: 'The average distance between crossing points is about 400 m, so the road is well served'). Her mean of 400 m is correct but incomplete. Learners return to their ungrouped table from Lesson 2: 11 different gap values among 20 gaps, from 100 m to 1,100 m (range 1,000 m from Lesson 3). They predict how the data could be made easier to read. Pairs then experiment with class widths (100 m, 200 m, 500 m), tally the 20 gaps, and build a grouped frequency table with classes 0 to <200, 200 to <400, 400 to <600, 600 to <800, 800 to <1000, 1000 to <1200 and frequencies 5, 9, 0, 2, 1, 3 (total 20). Pairs identify the modal class (200 to <400) and discuss what the empty class 400 to <600 and the right-hand cluster of 6 long gaps suggest. The DQB is

updated and the class model is revised: the grouped table shows the gaps bunched at the short end with a separate group of very long gaps.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Grouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Grouped Data to Find the Mean (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners open their Lesson 2 ungrouped frequency table of the 20 gaps on Thika Road (100:3, 150:2, 200:4, 250:3, 300:2, 600:1, 700:1, 800:1, 1000:1, 1050:1, 1100:1). In pairs they respond to: 'Ms Abdi says the average is 400 m. If you had to show the matatu drivers and parents the shape of all 20 gaps on one small table, what would you do to the 11 rows so that it is easier to read?' Each pair writes a prediction on a sticky note: how many groups they would use, and what size each group would be. They also predict which group would hold the most gaps.	Say: 'Last time we built a table with eleven different gap values. Look at it for 30 seconds. What do you notice about how it looks?' Hold WAIT TIME of 10-15 seconds before taking answers. Cold-call 4 pairs for their prediction of the number of groups and group size, and write each on the board without evaluating. Ask: 'Who thinks the biggest group will be around 100 m? Around 300 m? Around 900 m?' Take a show of hands. Prompt: 'Remember the range was 1,000 m from Lesson 3. How might that help you choose a group size?' Do not give the term 'class' or 'class width' yet; let learners use their own words first and then label them.	Prediction with public commitment (Predict-Observe-Explain). Learners commit to an idea about how to group before seeing the grouped table, so that later they can compare their prediction with the evidence and notice what their thinking missed. Introducing the word 'class' after their own wording connects everyday language to mathematical language.	Look for: pairs who mention the range of 1,000 m when choosing group size; pairs who propose equal-size groups versus uneven groups; pairs who predict that the largest group is at short gap values (100-399 m) because Lesson 2 showed values like 100, 150, 200, 250, 300 repeated. Record names of pairs who suggest groups that would overlap (for example 100-200 and 200-300), because the class boundaries misconception needs addressing in the Observe phase.
Observe Phase  VIDEO: Grouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Grouped Data to Find the Mean Practice <i>Related topic: not an exact match</i>	Pairs get the sorted list of 20 gaps (100,100,100,150,150,200,200,200,200,250,250,250,300,300,600,700,800,1000,1050,1100) and a blank tally sheet. Task 1 (8 minutes): try class widths 100 m, 200 m and 500 m. For each, count how many classes are needed to cover 100 m to 1,100 m and decide which is neither too crowded nor too thin. Task 2 (10 minutes): with width 200 m, start at 0 and write the classes as 0 to <200, 200 to <400, 400 to <600, 600 to <800, 800 to <1000, 1000 to <1200. Tally each gap into exactly one class (for example 200 m goes in 200 to <400, not in 0 to	Circulate and say: 'Show me where the gap of exactly 200 m goes. Why that class and not the one before it?' to reinforce that the lower boundary is included and the upper boundary excluded. Ask: 'With width 100 m, how many classes would we need? With width 500 m?' (11 classes; 3 classes) and 'Which one hides the pattern, and which one still leaves us with too many rows?' Give WAIT TIME of 10-15 seconds after each question. Stop the class at 14 minutes and cold-call 3 pairs for their frequencies; if any pair has a total other than 20, say: 'Your total is not 20. What does that tell you	Structured data collection and pattern noticing. Trying three class widths lets learners see for themselves how a choice changes what the data look like, and the tally forces them to treat each gap consistently. The empty class (400 to <600) is highlighted as a feature of the data, not an error.	Observable indicators: (1) tally for each gap sits in exactly one class and the frequencies are 5, 9, 0, 2, 1, 3 totalling 20; (2) gaps at exactly 200 m and 600 m are placed correctly in 200 to <400 and 600 to <800; (3) learners write classes with the '<' upper boundary' convention; (4) learners can justify 200 m as a sensible class width using the range of 1,000 m; (5) learners record 0 for the empty class 400 to <600 instead of leaving it blank. Note pairs who need support with boundaries and give them a number line from 0 to 1,200 marked every 200 m.

for this lesson. Source: CK-12 Search ARES for similar readings	<200). Complete the frequency column and check the total is 20. Achieng (data collection) reads values aloud, Kiprop (tables and calculations) records tallies and checks the total, Njeri (graphs) notes any class that looks surprising. Expected frequencies: 5, 9, 0, 2, 1, 3.	about your tallies? Recount with your partner.' Ask the whole class: 'Which class has the highest frequency? Hold up the class on your slate.' Introduce the term 'modal class' and the term 'class width' (200 m). Ask: 'Which class has frequency 0? What does that mean on the road?'		
Explain Phase  VIDEO: Grouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar videos  HTML: Grouped Data to Find the Mean (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Pairs use the completed grouped frequency table to discuss and identify the modal class (200 to <400, frequency 9). They then answer: (a) How many gaps are shorter than 400 m ($5 + 9 = 14$)? (b) How many gaps are 600 m or longer ($2 + 1 + 3 = 6$)? (c) What is the empty class and what does it tell us about the data (no gap between 400 m and 599 m, so the six long gaps are clearly separate from the others)? (d) Ms Abdi's mean of 400 m falls in the class 400 to <600, the class with no gaps. What does that tell us? Pairs write a two-sentence explanation: 'The modal class is _____. This tells us _____ but it does not tell us _____.' Then the class discusses: 'The officer's claim is true but incomplete. Which feature of the grouped table shows what the mean hides?'	Say: 'The modal class is 200 to <400. Turn to your partner: does that mean the road is well served everywhere? Explain in one minute.' Hold WAIT TIME of 10-15 seconds, then cold-call 4 pairs. Press for evidence: 'Point to the row in your table that supports your claim.' Ask: 'The average of 400 m sits in a class with zero gaps. Why might that be a warning sign about relying on the average?' Support with a number line on the board showing 14 gaps clustered at the left and 6 spread to the right. Say: 'The modal class tells us the most common gap size, but it does not show where along the road the long gaps are. What other information do we need?' (Position of gaps along the road and pedestrian counts.) Do not calculate grouped mean or median; say: 'We will use these classes next lesson to estimate the mean and median.' Reinforce respectful language: 'Ms Abdi's figure is correct; our table adds to it.'	Claim-Evidence-Reasoning with a Pair-Share and Cold Call. Learners make a claim about what the modal class shows, cite specific rows of the table as evidence, and explain why a single summary (the modal class or the mean) cannot locate risk. Contrasting the modal class with the empty class and the long-gap cluster builds the idea that distributions have shape.	Look for: correct identification of the modal class as 200 to <400 with frequency 9 (not '9' as the class and not '300'); statements that 14 of 20 gaps are under 400 m and six gaps are 600 m or longer; recognition that the modal class describes the typical short gap but does not show the long gaps; use of evidence such as 'the 1000 to <1200 class has 3 gaps'. Misconceptions to catch: confusing the modal class with the class with the largest values; confusing frequency with class; reading the mean as a class that must contain data.
Driving Question Board (DQB) Creation  VIDEO: Grouped Data to Find the Mean Principles	Learners review the DQB. In pairs they add one new sticky note under each of the three KICD questions: What is statistics? How do we represent data? How do we use statistics in day-to-day life? Examples: 'Why did we choose 200 m and not 100 m or 500 m?'	Say: 'Write one question that today's table has raised for you. Put it under the key inquiry question it fits best.' WAIT TIME of 10-15 seconds, then cold-call 3 learners to read their stickies aloud and say where they fit. Ask: 'Which of our earlier questions did the	Driving Question Board as a living record of understanding. Sorting questions under the three KICD inquiry questions shows how today's table fits into the bigger purpose, and colour-coding answered questions makes learning progress visible and	Indicators: each pair adds at least one question that links grouping to the road (not only to procedure); at least two questions anticipate Lesson 5 (grouped mean and median) or Lesson 6 (histogram); learners place questions under the correct KICD inquiry question.

<i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Grouped Data to Find the Mean Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	'Can two groups choose different classes and get different shapes?' 'Which gaps in the 1000 to <1200 class are near the school gate?' 'How do we find the mean and median if the table only has classes?' They place stickies and mark questions answered today (How do we represent data? grouping into classes and the modal class) in a different colour.	grouped table help us answer? Which new question will Lesson 5 begin to answer?' Circle 'How do we find the mean and median if the table only has classes?' and say: 'That is our next step.' Link back to the phenomenon: 'Ms Abdi said the road is well served and Mzee Otieno says the near-misses are in one stretch. What does our table add to this argument?'	motivates the next lesson.	Note learners whose questions only restate 'what is a class?' as needing a short follow-up with the table.
Model Building Phase  VIDEO: Grouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Grouped Data to Find the Mean (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners revise the class model of the phenomenon begun in Lesson 1. On a poster headed 'Why does an average of 400 m not tell the whole story?' each pair adds a new panel: a small grouped frequency table of the 20 gaps (classes 0 to <200 to 1000 to <1200 with frequencies 5, 9, 0, 2, 1, 3) and a quick sketch of a bar for each class, with the modal class 200 to <400 marked. Under it they write: 'Most gaps are short (14 of 20 are under 400 m), but six gaps are 600 m or longer and none lie between 400 and 600 m. The mean of 400 m sits in the empty class. The modal class does not show where the long gaps are.' They add a question mark beside 'Where are the long gaps along the road and how many pedestrians cross there?' for later lessons.	Say: 'Add a panel to our model that a Parents Association member could understand in 30 seconds.' Circulate and ask: 'Where does the officer's mean fit on your sketch? Is it in a class with gaps?' Give WAIT TIME of 10-15 seconds. Cold-call 3 pairs to share their panel and ask the others: 'What did they show well? What would you add?' Remind: 'The officer's claim is true but incomplete. Our grouped table shows part of the story: the gaps are bunched short, with a separate long group.' Ask them to keep the poster for Lesson 5 where they will add the grouped mean and median. Do not draw histograms yet; sketches are rough bars only.	Cumulative model revision (Gallery-style sharing). Learners add a new representation to the evolving model and explain it in their own words. Sharing with peers lets them compare how different pairs describe the same table and refine the language of 'classes', 'frequency' and 'modal class'.	Observable indicators: the panel shows all six classes with frequencies 5, 9, 0, 2, 1, 3 and a total of 20; the modal class 200 to <400 is correctly marked; the written statement mentions both short gaps (14 of 20 under 400 m) and long gaps (six of 600 m or more); the empty class 400 to <600 is noted; learners use 'estimate-ready' language (for example 'we will use the classes to estimate the mean next lesson'). Use a quick exit slip: 'Name the modal class and say one thing it cannot tell us about Thika Road.'

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did every pair place boundary values (200 m, 400 m, 600 m, 1000 m) in the correct class, and which pairs needed the number line? (2) Could learners justify the 200 m class width using the 1,000 m range from Lesson 3, or did they only follow the instruction? (3) Did learners treat the empty class 400 to

<600 as meaningful data and notice that the mean of 400 m falls inside it? (4) Did any learner confuse the modal class with the class holding the largest values, or the frequency 9 with the class? (5) Did pairs say that the modal class does not show where the long gaps are along the road, and did the language stay respectful of Ms Halima Abdi's correct but incomplete figure? (6) Which DQB questions will you pick up in Lesson 5 (grouped mean and median) and Lesson 6 (histograms)? Plan extra support for pairs who struggled with tallies and consider pairing them with learners who explained boundaries well.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the 20 gaps on the 8,000 m stretch of Thika Road, which have 11 different values from 100 m to 1,100 m, can be grouped into six classes of width 200 m: 0 to <200 (5 gaps), 200 to <400 (9 gaps), 400 to <600 (0 gaps), 600 to <800 (2 gaps), 800 to <1000 (1 gap) and 1000 to <1200 (3 gaps), with a total of 20.
What did I learn?	We learned how to choose a class width using the range, write classes with the lower boundary included and the upper excluded (for example 200 to <400), tally each value into exactly one class, check that the frequencies add up to 20 and identify the modal class as the class with the highest frequency, which is 200 to <400.
How does this explain the phenomenon?	The grouped table shows that 14 of the 20 gaps are under 400 m while six gaps are 600 m or longer, and no gap lies in 400 to <600 where the officer's mean of 400 m falls. So the officer's claim is true but incomplete: the modal class shows the most common gap but cannot show where along Thika Road the long gaps and the risk are.

LESSON 5 (80 minutes (double lesson)): Mean and median from grouped data: what happens to the average when we group the gaps?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners estimate the mean and median of the grouped Thika Road gap data using class midpoints and cumulative frequency, compare the estimates with the ungrouped results (400 m and 250 m), and use the comparison to revise the class model of why one average cannot show where pedestrians are at risk.
Knowledge	By the end of the lesson the learner should be able to determine the mean and median of grouped data (KICD Specific Learning Outcome c: determine mean, mode and median of grouped and ungrouped data). Learners know that midpoint = (lower + upper)/2, that the grouped mean = sum of (f x midpoint)/sum of f is an estimate, and that the grouped median = $L + ((n/2 - cf)/f) \times w$. Learners know that grouping loses detail, so estimates differ from the ungrouped values.
Skills	Calculating midpoints; completing an f x midpoint column and summing it; building a cumulative frequency column; locating the median class from n/2; substituting into the median formula; comparing estimates with true values; Critical thinking and problem solving (NGSS SEP: Analyzing and interpreting data; Using mathematics and computational thinking).
Attitudes	Integrity: learner records and calculates accurately and reports differences between estimates and true values honestly. Unity: learner works in groups, sharing roles (Achieng data, Kiprop calculations, Njeri graphs). Love: learner is considerate of group members as they check each other's working. Life skills (Decision Making): learner sees why decisions about road safety need more than one average.
Key Inquiry Question	KICD key inquiry questions: 1. What is statistics? 2. How do we represent data? 3. How do we use statistics in day-to-day life? This lesson advances Q2 and Q3: How does grouping change the averages we calculate, and can we trust an estimate when making decisions about a road?
Purpose in Storyline	Lesson 3 gave the ungrouped mean (400 m) and median (250 m); Lesson 4 built the grouped table with classes of width 200 m (frequencies 5, 9, 0, 2, 1, 3). Lesson 5 uses that table to estimate the mean (440 m) and median (311 m) and compare them with the ungrouped values. Learners see that grouping gives estimates close to but not equal to the true values, and that both the mean and the median still sit far from the long 600 m to 1,100 m gaps in the long stretch G9-G14. The comparison feeds the class model and prepares for the histograms in Lesson 6.
Safety Notes	This is a classroom calculation lesson. Any data collection or road observation is NOT done in this lesson; use the recorded Thika Road data only. If learners discuss the road, remind them never to cross at undesignated spots, and to cross only at zebra crossings or footbridges. Keep walkways between desks clear while groups move to the board.

B. LESSON OVERVIEW

Ms Halima Abdi's claim of a 400 m average gap is arithmetically true, but learners have seen that 14 of the 20 gaps are 300 m or shorter while six long gaps cluster around the school gate. In Lesson 4 the class grouped the 20 Thika Road gaps into six classes. Today learners first predict whether grouping will change the mean and median. They then complete the grouped table for the 20-gap sample: midpoints (100, 300, 500, 700, 900, 1100), f x midpoint (500, 2700, 0, 1400, 900, 3300; sum 8,800) and an estimated mean of $8,800/20 = 440$ m. Next they build cumulative frequencies (5, 14, 14, 16, 17, 20), find the median class (200 to <400) and use the formula to get $200 + ((10-5)/9) \times 200 = 311$ m. They compare 440 m and 311 m with the true 400 m and 250 m and explain why grouping makes the estimates differ. Finally they add a DQB question and revise the class model: the officer is right about the mean, but the mean and median are only summaries, and an estimate from grouped data is still far from showing the danger in the long stretch.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Grouped Data to Find the Mean Principles Source: CK-12  Search ARES for similar videos  HTML: Grouped Data to Find the Mean (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Learners recall from Lessons 3 and 4: the ungrouped Thika Road gap mean was 400 m and the median 250 m, and the 20 gaps are now sorted into six classes of width 200 m with frequencies 5, 9, 0, 2, 1, 3. In pairs they individually write a prediction on a sticky note or in their books: 'If we calculate the mean and median from the grouped table instead of the 20 original gaps, will we get 400 m and 250 m exactly, a bit higher, or a bit lower? Why?' They also guess which class the median will fall in. Pairs then share predictions with a neighbouring pair and record a group prediction on a prediction chart.	Display the grouped table on the board (classes 0 to <200 ... 1000 to <1200 with 5, 9, 0, 2, 1, 3). Say: 'Ms Halima Abdi's mean of 400 m is correct for the 20 original gaps. Now I hide the original gaps and only give you this table. What do you think will happen to the mean and the median?' Give 1 minute silent writing, then WAIT TIME 10-15 seconds after asking: 'Which class do you think the 10th and 11th gaps fall in?' Cold-call 4 learners from different groups to share predictions and reasons without saying whether they are right. Circulate and note ideas such as 'grouping loses the exact values' or 'the answer will be exactly the same'. Record the 3 most common predictions on the board and tell learners we will test them with the data.	Prediction chart and Think-Pair-Share: writing a prediction before calculating commits learners to an idea, so that later differences between 400 m / 250 m and 440 m / 311 m become a puzzle to explain rather than just a number to copy. Sharing with another pair surfaces the misconception that grouped and ungrouped answers must be identical.	Look for: (1) at least 80% of learners have written a prediction with a reason before the share; (2) learners can state that the grouped table no longer shows the individual gaps (100 m, 150 m, ... 1,100 m); (3) note how many predict the grouped mean will be exactly 400 m. Use a thumbs show: up = same, sideways = close but different, down = very different, and record the counts on the board for return in the Explain phase.
Observe Phase  VIDEO: Grouped Data to Find the Mean Principles Source: CK-12  Search ARES for similar videos  EXERCISE: Grouped Data to Find the Mean Practice Source: CK-12  Search ARES for similar readings	(25 min) In groups of 4 (Achieng, Kiprop, Njeri and one checker per group) learners work with the Thika Road grouped gap table for the 20-gap sample. PART A, mean: they add a midpoint column using midpoint = (lower + upper)/2, getting 100, 300, 500, 700, 900, 1100; they add an f x midpoint column: 5 x 100 = 500, 9 x 300 = 2700, 0 x 500 = 0, 2 x 700 = 1400, 1 x 900 = 900, 3 x 1100 = 3300; they sum to 8,800 and divide by 20 to get an estimated mean of 440 m. PART B, median: they add a cumulative frequency column (5, 14, 14, 16, 17, 20), find $n/2 = 10$, see that the 10th value lies in the	Distribute the grouped table and calculators. Say: 'Njeri, Kiprop and Achieng, share the jobs: one reads the frequency, one calculates, one checks. Swap after Part A.' Model only the first row aloud: 'The class 0 to <200 has lower boundary 0 and upper boundary 200, so midpoint is $(0 + 200)/2 = 100$. Why do we use one value to stand for the whole class?' Then release groups. Circulate with a checklist; ask groups: 'Why is f x midpoint for the class 400 to <600 zero?' and 'Why does the cumulative frequency stay 14 in the third row?' After Part A, stop the class and ask 3 groups to report their sum	Structured data table with role rotation (Analysing and interpreting data; Using mathematics and computational thinking): building the midpoint, f x midpoint and cumulative frequency columns step by step makes the logic of the estimate visible. Using one number to stand for each class helps learners see where the approximation comes from. Placing the grouped and ungrouped results side by side creates the evidence needed for the Explain phase.	Observable indicators: (1) correct midpoints 100, 300, 500, 700, 900, 1100 in the table; (2) correct f x midpoint values and sum of 8,800; (3) estimated mean 440 m written with the unit and the word 'estimate'; (4) cumulative frequencies 5, 14, 14, 16, 17, 20; (5) the median class 200 to <400 chosen because $n/2 = 10$; (6) correct substitution $L = 200$, $cf = 5$, $f = 9$, $w = 200$ and the answer 311 m. Common errors to note: using class boundaries instead of midpoints, dividing by 6 (the number of classes) instead of 20, using f instead of cumulative frequency for cf, using $n = 20$

	class 200 to <400 (cf before it = 5, f = 9, L = 200, w = 200) and calculate $200 + ((10 - 5)/9) \times 200 = 311$ m (311.1, to the nearest metre). They record both estimates on a results card and place the results beside the ungrouped values: mean 400 m and median 250 m.	(8,800) and mean (440 m); WAIT TIME 10-15 seconds before taking answers so slower groups can recheck. Before Part B ask: 'Where in the cumulative column do we first reach or pass 10?' Cold-call 3 learners. Write the formula $L + ((n/2 - cf)/f) \times w$ on the board and label each symbol with an arrow to the table. Close with: 'Place your two estimates next to 400 m and 250 m. What do you notice?'		instead of $n/2 = 10$. Rubric alignment: Meets = accurate table and both estimates; Approaches = needs help with one of the two; Below = partially completes the table.
Explain Phase  VIDEO: Statistics intro: Mean, median, & mode <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Grouped Data to Find the Mean (Flexbook) Source: CK-12  Search ARES for similar readings	(20 min) Groups compare their grouped estimates with the ungrouped results on a comparison chart: mean 400 m vs 440 m (grouped estimate, 40 m higher); median 250 m vs 311 m (grouped estimate, 61 m higher). They answer in their groups: (1) Why does the grouped mean differ from 400 m? (Midpoints replace the real gaps; e.g. the 9 gaps in 200 to <400 are treated as 300 m, although the real values are 200, 250 and 300 m, so the class midpoint is higher than the real gaps; the six values in 600 to <1200 are treated as midpoints too, so the total changes.) (2) Why is the grouped median 311 m while the true median is 250 m? (The formula assumes the 9 gaps in 200 to <400 are spread evenly, but in fact they are bunched at 200 m to 300 m.) (3) Which estimate is closer to the truth, and does it change what we can say about the officer's claim? Learners then connect to the phenomenon: both the mean and the median, ungrouped or grouped, describe a typical gap of 250 m to 440 m, yet the six gaps of 600 m to 1,100 m sit in the long stretch G9-G14 that includes the school gate, where	Ask: 'Our predictions on the board said same, close, or very different. Which was right?' Use the thumbs counts from the Predict phase. Pose Question 1 and WAIT TIME 10-15 seconds. Use talk moves: 'Who can add to what Achieng said?' and 'Can someone say that in a different way?' Cold-call 4 learners (one per question stage) and ask them to point to the table when explaining. Use the Kenyan analogy: 'If you only know that a pupil scored between 40 and 60 in Mathematics at Kisumu, you use 50 for everyone. Is that exact?' Then ask: 'The officer says 400 m; our grouped mean says 440 m; the median says 250 m or 311 m. Is the officer wrong?' Guide learners to say: 'The officer's claim is true but incomplete.' Ask: 'If a matatu driver like Mzee Otieno asks where the danger is, can any of these four numbers tell him?' Take 3 responses, and guide to: grouping gives estimates; none of the averages tells us WHERE the long gaps are.	Claim-Evidence-Reasoning with a comparison chart: learners make a claim (the grouped estimates are close but not exact), support it with the figures 400 vs 440 and 250 vs 311, and give reasoning about midpoints and even spread within classes. The second reasoning step connects the mathematical idea (grouping loses information) to the phenomenon (an average hides the long gaps), which advances the driving question.	Listen for: (1) learners state that the grouped mean is an ESTIMATE, not equal to 400 m; (2) learners explain the difference using midpoints and the assumption of even spread within a class; (3) learners say the officer's claim is true but incomplete, not wrong; (4) learners link to the long stretch G9-G14 and the 140 of 182 pedestrians. Check with an exit prompt on mini-whiteboards: 'Write one reason the grouped median (311 m) is not 250 m.' Scan for 'spread evenly' or 'we lost the real values'. Note learners who say the grouped answer is 'wrong' and revisit: it is an estimate, useful when only grouped data are available.

	140 of 182 pedestrians crossed at undesignated spots in the one-morning sample.			
Driving Question Board (DQB) Creation  VIDEO: Analyzing a cumulative relative frequency graph <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Grouped Data to Find the Mean Practice Source: CK-12  Search ARES for similar readings	(10 min) Learners return to the class DQB started in Lesson 1. Individually, they write on sticky notes: one thing they now understand (for example 'How to find the estimated mean from midpoints') and one new question (for example 'Can a graph show us where the long gaps are?', 'Why is the grouped median 311 m and not 250 m?', 'Would smaller classes make the estimate closer?'). They place the notes under the three KICD key inquiry questions (What is statistics? How do we represent data? How do we use statistics in day-to-day life?) and mark any earlier questions they can now answer, such as 'How can the mean be correct but the road still dangerous?' (partly answered: the mean is pulled up by a few very long gaps).	Say: 'Let us look at our board. Which of our earlier questions can we move to Answered or Partly answered?' Give WAIT TIME 10-15 seconds, then cold-call 3 learners to place notes and justify them. Ask Njeri's group: 'Your question about graphs - which lesson might help?' (preview Lesson 6: histograms). Group similar questions with learners' help. Highlight a question about class width and flag it as a good question for Lessons 6 and 7. Close: 'The grouped mean and median are estimates, but they still do not tell Mzee Otieno where the danger is. What do we need next?'	Driving Question Board as a public record of growing understanding: learners see their questions move from unanswered to partly answered, which makes progress visible and keeps the investigation anchored in the phenomenon. Sorting under the three KICD key inquiry questions links the work to curriculum goals.	Check that every learner has contributed at least one note; that at least two new questions refer to graphs or to showing WHERE the long gaps are; and that learners correctly place calculation notes under 'How do we represent data?' or 'How do we use statistics in day-to-day life?'. Note learners whose questions repeat Lesson 1 questions without any new content, and prompt them with 'What did the table show that you could not see before?'.
Model Building Phase  VIDEO: Grouped Data to Find the Mean Principles Source: CK-12  Search ARES for similar videos  EXERCISE: Estimating mean and median in data displays <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)	(15 min) Groups revise their class model (a poster or chart begun in Lesson 1) of why the officer's average does not show where pedestrians are at risk. They add a new section titled 'What grouped averages tell us', containing: the grouped table; the estimated mean 440 m; the grouped median 311 m; the ungrouped mean 400 m and median 250 m; and a short statement, for example: 'Mean (400 m) and median (250 m) describe a typical gap on this 8,000 m stretch of Thika Road in our 20-gap sample. Grouping gives estimates (440 m and 311 m) that are close but not exact. None of these averages shows	Say: 'Add to your model what today's numbers show and what they still cannot show.' Prompt: 'Use the words estimate, midpoint and cumulative frequency in your model.' Circulate and ask: 'Does your model say the officer is wrong? How could you word it so it is fair to Ms Halima Abdi?' Remind groups that the claim is true but incomplete. Give WAIT TIME 10-15 seconds after asking: 'If a learner from Mombasa Road gave you only a grouped table, what could you still work out?' Run the gallery walk: each group leaves one person at the poster while others visit; cold-call 3 visitors to report one idea they borrowed.	Cumulative model revision with gallery walk: adding today's evidence to the existing model shows learners that scientific and mathematical models grow as evidence grows. Writing what the model still cannot explain creates a purposeful need for histograms and polygons in the next lessons, linking directly to the driving question about showing where pedestrians are really at risk.	Look for: (1) the model includes the grouped mean 440 m and grouped median 311 m labelled as estimates; (2) it compares them with 400 m and 250 m; (3) it states a limit: averages do not show where the long gaps are; (4) the wording is respectful: true but incomplete; (5) a next step (graphs) is named. Rubric check: Meets = accurate figures with an explanation; Exceeds = also explains why grouping causes the difference; Approaches = figures with minimal explanation; Below = figures copied without explanation.

Search ARES for similar readings	that six gaps of 600 m to 1,100 m are in one stretch.' Groups then add a 'What we still need' arrow pointing to graphs (Lesson 6: histograms; Lesson 7: frequency polygons) and a note that this is a one-morning sample and should be repeated. Each group shares its model addition in a gallery walk.	Finish by asking 2 learners to complete the sentence 'Averages alone cannot show risk because ...'.		
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D. TEACHER REFLECTION

After the lesson ask yourself: Did every learner correctly find midpoints and complete the $f \times \text{midpoint}$ column, or did some use class boundaries or divide by 6 instead of 20? Could learners explain WHY the grouped mean (440 m) and median (311 m) differ from 400 m and 250 m, or only that they do? Did anyone call the grouped answer 'wrong'? Did the discussion keep the officer's claim as true but incomplete? Which groups struggled with $n/2$ and cumulative frequency in the median formula, and what extra practice (for example a small table of 10 matatu fares from Kisumu to Kakamega) is needed? Did the learners link the averages back to the long stretch G9-G14 and the school gate? Which DQB questions should shape Lesson 6?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We used the grouped Thika Road gap table for the 20-gap sample (classes 0 to <200 ... 1000 to <1200, frequencies 5, 9, 0, 2, 1, 3) to calculate class midpoints (100, 300, 500, 700, 900, 1100), $f \times \text{midpoint}$ (sum 8,800) and cumulative frequencies (5, 14, 14, 16, 17, 20). We found an estimated mean of 440 m and a grouped median of 311 m, and compared them with the ungrouped mean (400 m) and median (250 m).
What did I learn?	For grouped data, the mean is estimated using class midpoints: sum of ($f \times \text{midpoint}$) divided by sum of f . The median is found with cumulative frequency and the formula $L + ((n/2 - cf)/f) \times w$. Grouping loses the exact values, so grouped results are estimates that differ from ungrouped results.
How does this explain the phenomenon?	The grouped mean (440 m) and median (311 m) differ from 400 m and 250 m because midpoints stand in for the real gaps and the formula assumes values are spread evenly in a class. Ms Halima Abdi's 400 m mean is true but incomplete: none of these averages shows that six gaps of 600 m to 1,100 m sit in one 5,250 m stretch near the school gate, so we need graphs to show where pedestrians are at risk.

LESSON 6 (40 minutes (one lesson; extend to a double period if learners draw both histograms in full)): Histograms and frequency density: seeing where the long gaps are on Thika Road

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners draw a histogram of the 20 Thika Road gaps first with equal class widths and then with unequal class widths using frequency density, and use the bars to see where the long gaps appear, so they can show why the officer's mean of 400 m is true but incomplete.
Knowledge	By the end of the sub-strand the learner should be able to: d) represent data using histograms and frequency polygons; e) interpret data from histograms and frequency polygons. In this lesson learners draw histograms (with equal and unequal class-width) from collected data. Equal-width classes (width 200 m) for the 20 gaps on the 8,000 m Thika Road stretch have frequencies 5, 9, 0, 2, 1, 3. For unequal widths: 0 to <200 (f=5), 200 to <400 (f=9), 400 to <1000 (f=3), 1000 to <1200 (f=3). Frequency density = frequency / class width (in units of 100 m), giving 2.5, 4.5, 0.5, 1.5. Bars touch, the horizontal axis is continuous, and bar area represents frequency.
Skills	Calculating frequency density; choosing and labelling scales on both axes; drawing bars that touch with correct boundaries; reading the shape of a histogram; explaining why the tallest bar must be judged by frequency density, not frequency, when widths are unequal; using and interpreting graphs as models (NGSS SEP: Developing and Using Models; Analyzing and Interpreting Data; Constructing Explanations).
Attitudes	Core competencies: Critical thinking and problem solving (the learner uses the histograms to examine data and make conclusions); Communication and collaboration. Values: Integrity (the learner records and plots data accurately); Unity (the learner works in groups to draw histograms). PCIs: Life Skills (Decision Making); Road safety. Learners are respectful of the officer, whose mean is true but incomplete.
Key Inquiry Question	How do we represent data? (Also tracked: What is statistics? How do we use statistics in day-to-day life?) Lesson question: What does a histogram of the 20 Thika Road gaps show that the mean of 400 m does not, and why must we use frequency density when class widths are unequal?
Purpose in Storyline	Lessons 4 and 5 gave a grouped table (frequencies 5, 9, 0, 2, 1, 3) and showed that grouping changes estimates (grouped mean 440 m versus true 400 m). Lesson 6 turns the table into a picture. For the first time the class can SEE the cluster of short gaps and the separate long gaps of 600 m to 1,100 m that make up the long stretch G9-G14, which includes the school gate. This is a key piece of evidence for the final model: one average hides the shape of the data. Lesson 7 then draws frequency polygons and compares Thika Road with Mombasa Road.
Safety Notes	The lesson is classroom-based, so no learners go onto the road. If the class revisits the road to check any measurement, this must be done only with Mwalimu Wanjiru and other adults, on the school side of the road, using the designated crossing. Learners must not run across Thika Road to measure anything. Pencils and rulers are used sensibly. Graph paper and charts are shared fairly.

B. LESSON OVERVIEW

Mwalimu Wanjiru reminds the class of the officer's claim: the average gap between the 21 designated crossing points on the 8,000 m stretch of Thika Road is about 400 m. Ms Halima Abdi's arithmetic is correct, yet Mzee Otieno's matatu near-miss log has 9 of 12 near-misses in one stretch. Learners first predict what a picture of the 20 gaps will look like. They then draw a histogram from the equal-width table (frequencies 5, 9, 0, 2, 1, 3) and notice the empty 400 to <600 bar and the lonely bars to the right. Next the teacher merges the classes 400 to <600, 600 to <800 and 800 to <1000 into one class, 400 to <1000. Learners find that using frequency as the bar height makes this wide class look misleadingly tall. They calculate frequency density (2.5, 4.5, 0.5, 1.5), redraw, and discover that area, not height, represents frequency. In the

Explain phase they use both histograms to say where the long gaps are and why the mean of 400 m looked reassuring. The class updates the DQB and revises the model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Density Curves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Applications of Histograms Practice Source: CK-12  Search ARES for similar readings	Learners recall the officer's claim and Mzee Otieno's remark. Each learner sketches, in their exercise book, what they think a bar graph of the 20 Thika Road gaps will look like (short, medium and long gaps on the horizontal axis) and writes one sentence: 'I think the picture will show...' They compare with a partner. A second prediction: if we merge three classes into one wide class, will the bar be taller or shorter? Why?	Display on the board: 'Officer: the average gap is about 400 m, so the road is well served. Mzee Otieno: most near-misses are in one stretch.' Ask: "If I drew all 20 gaps as a picture, where would you expect the tall bars? Sketch it silently for 2 minutes." Give WAIT TIME of 10-15 seconds after asking the question before any learner speaks. Circulate and select 3 contrasting sketches. Cold-call 3 learners (for example Achieng, Kiprop, Njeri's group or others) to describe their sketch without saying whether it is right. Then ask: "If the bars have the same width, what does the height tell us? If one bar is three times wider, what do you think should happen to its height?" Record 2 predictions on the board without evaluating them.	Predict-Observe-Explain (prediction stage) with Public Sketch Comparison: committing to a picture before drawing exposes the common idea that 'average = typical' and gives learners something to test against the real histogram. Selecting contrasting sketches makes disagreement visible and productive.	Look for: (1) whether sketches place a single hump near 400 m (the mean misconception) or show short gaps plus a tail; (2) whether learners predict that a wider bar should be shorter, or just assume height equals frequency. Tally on the board how many learners predict a hump near 400 m. Use this tally later to compare with the real histogram.
Observe Phase  VIDEO: Histograms <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Applications of Histograms	In groups of 4 (roles: Achieng-type recorder of tallies, Kiprop-type table and calculation checker, Njeri-type grapher, one reader/checker), learners use graph paper. Part A (equal widths): from the grouped table (0 to <200: 5; 200 to <400: 9; 400 to <600: 0; 600 to <800: 2; 800 to <1000: 1; 1000 to <1200: 3) they draw a histogram with the horizontal axis as gap length in metres (continuous, labelled 0, 200, 400... 1200), vertical axis frequency, bars touching, and a gap with no bar for	Model the first bar on the board: "The first class is 0 to <200, so the bar goes from 0 to 200 on the axis and its height is 5." Say: "Bars touch because the gap length is continuous. Why is there no break between 200 and 400?" Allow WAIT TIME of 10-15 seconds. For Part B, say: "Our wide class 400 to <1000 holds 3 gaps: 600 m, 700 m, 800 m. Draw it with height 3 first. What does your picture suggest about gaps between 400 m and 1000 m compared with 200 to <400?" Cold-call 2 groups to	Productive Struggle with Cognitive Conflict: the frequency-height drawing of the wide class creates a visible conflict (it seems to say gaps of 400 m to 1000 m are common), and frequency density resolves it. Gallery Walk and Modelling: learners compare drawn graphs, building an abstract idea (area represents frequency) from concrete bars.	Observable indicators: (1) bars touch and have correct boundaries on the continuous axis; (2) the 400 to <600 equal-width bar is correctly shown as zero height; (3) frequency densities calculated correctly as 2.5, 4.5, 0.5, 1.5; (4) the wide bar 400 to <1000 is drawn at height 0.5, the lowest bar; (5) axes labelled with 'Gap length (m)' and 'Frequency' or 'Frequency density'. Use a checklist while circulating; note groups who divide by 600 instead of 6 (the unit error) and prompt:

Practice Source: CK-12 Search ARES for similar readings	400 to <600 (frequency 0). Part B (unequal widths): the teacher gives the new table: 0 to <200 (f=5), 200 to <400 (f=9), 400 to <1000 (f=3: gaps 600, 700, 800), 1000 to <1200 (f=3). Groups first draw bars with height = frequency and notice that the wide class looks as tall as other bars, and that the picture changes the story. They then calculate frequency density = frequency / class width in units of 100 m: $5/2 = 2.5$, $9/2 = 4.5$, $3/6 = 0.5$, $3/2 = 1.5$, and redraw using frequency density on the vertical axis. They shade or count squares to check that bar area represents frequency.	share what they see. Then ask: "Is this fair? The class is three times as wide. How could we make the bar height fair?" Introduce frequency density only after learners voice the problem: "Frequency density = frequency divided by class width, with width in units of 100 m." Circulate, check scales and that each group has 2.5, 4.5, 0.5 and 1.5. Ask one group to count squares to show area = frequency (for example the first bar: 2.5 high x 2 wide = 5). Provide extra support cards for learners needing scaffolding (for example pre-labelled axes).		"What unit are we using for class width?" Exit-check: 3 cold-called learners state the units.
Explain Phase  VIDEO: Density Curves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Represent Real-World Data Using Bar Graphs, Frequency Tables and Histograms (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners use both histograms to explain the phenomenon. In groups they answer: (1) In the equal-width histogram, where are the tall bars and where are the lonely bars? (2) Why is the bar for 400 to <1000 the lowest in the unequal-width histogram although it holds 3 gaps, the same as 1000 to <1200 (also 3 gaps)? (3) Fourteen of the 20 gaps are 300 m or shorter and six gaps are 600 m to 1,100 m: where do these six gaps lie on the road (G9-G14, the long stretch from the 1,550 m mark to the 6,800 m mark, including the school gate at the 4,000 m mark)? (4) How does this show why the mean of 400 m is true but incomplete? They write a 3-sentence explanation: 'The histogram shows... The mean hides this because... Pedestrians are at risk in...' Link to pedestrians: 140 of the 182 undesignated crossers (77%) and 9 of 12 near-misses (75%) are in the long stretch.	Ask: "Look at your histogram. Where is the tall bar, and what does it tell you about most gaps on Thika Road?" WAIT TIME 10-15 seconds, then cold-call 4 learners. Press for evidence: "Which bar shows the gaps of 1,000 m, 1,050 m and 1,100 m? Which gaps on the road are those?" Then: "The officer's mean is 400 m. Where is 400 m on your histogram? Is it where most of the gaps are?" Guide learners to see that few or no gaps are actually near 400 m (the 400 to <600 bar is empty). Say: "Ms Halima Abdi is not wrong; her mean is correct but incomplete. What does the picture add?" Ask learners to explain frequency density to a partner in one sentence: "A tall bar of high density means gaps packed closely in that range, not just many gaps." Cold-call 3 pairs. Correct the misconception that the tallest bar is always the highest frequency when widths are unequal. Remind that the tallest bar is the highest frequency	Claim-Evidence-Reasoning (CER) with Talk Moves: learners make a claim about where the danger lies, support it with bars from the histogram, and reason using frequency density. Revoicing and 'add on' talk moves build a shared explanation that connects the graph to the phenomenon.	Listen for: (1) learners reading the shape correctly (a cluster of short gaps and a separate group of long gaps to the right); (2) using 'frequency density' and 'class width' accurately; (3) linking the long gaps to the long stretch G9-G14 and the school gate with the numbers 5,250 m, 140 of 182 pedestrians. Collect the 3-sentence explanations and sort into: describes shape only / links to the mean / links to pedestrian risk. Plan a short revisit of any group still saying the tallest bar is always the most frequent.

		density. Do not re-teach mean or median calculation (done in Lessons 3 and 5).		
Driving Question Board (DQB) Creation  VIDEO: Histograms <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Frequency Tables and Histograms (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the class DQB. They add sticky notes: new answers or evidence for the key inquiry questions (What is statistics? How do we represent data? How do we use statistics in day-to-day life?) and new questions arising, such as 'How can we show the gap pattern as a line graph to compare Thika Road with Mombasa Road?' or 'Which crossing points should be added first?' They move or mark questions answered by today's histograms, for example 'How do we represent data?' now has the evidence 'histograms with equal and unequal widths, using frequency density'.	Say: "Take 2 minutes silently to write one thing the histogram taught you and one new question. Then place your note on the DQB." WAIT TIME 10-15 seconds before inviting the first learner. Cold-call 3 learners to read a note and say which key inquiry question it connects to. Ask: "Which question on our board did today's evidence help us answer? Which still needs data?" Point out that Lesson 7 will draw frequency polygons and compare Thika Road with Mombasa Road. Group similar questions and highlight one that connects to the community presentation in Lesson 8.	Driving Question Board as a public record of learning: sorting questions into answered, partly answered and open shows learners how today's evidence moves the class understanding forward and sustains motivation for the next lesson.	Check that notes are specific (for example mention frequency density or the wide class) rather than generic. Count how many learners connect a note to one of the 3 KICD key inquiry questions. Note learners whose questions still reflect the 'mean = safe' idea and plan to address them in Lesson 7 and Lesson 8.
Model Building Phase  VIDEO: Density Curves <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Frequency Tables and Histograms Practice	Learners revise the class model of the phenomenon (a poster or diagram on a chart). They add a sketch of the histogram and annotate: 'Most gaps are short (14 of 20 are 300 m or shorter). A few very long gaps (600 m to 1,100 m) sit to the right and pull the mean up to 400 m. These long gaps are in the long stretch G9-G14, 5,250 m, with the school gate.' They add the rule: 'With unequal class widths, compare frequency density, not frequency.' They then complete their summary table row for this lesson.	Ask: "What can we now add to our model that we could not show in Lesson 5?" WAIT TIME 10-15 seconds. Cold-call 3 learners to propose one addition each and ask the class to agree or add on. Prompt: "Which part of the histogram best shows the danger near the school gate? What would you say to Ms Halima Abdi and Mzee Otieno using this graph?" Remind learners that this is a one-morning sample (Tuesday, 6:30-7:30 am) and should be repeated. Link ahead: "Next lesson we will draw frequency polygons and compare with Mombasa Road." Collect one model annotation per	Cumulative Model Revision: adding the histogram to the model makes the graph a tool for explanation, not just a drawing. Learners must say what the model can now explain (where the long gaps are) and what it cannot yet (comparison with another road).	Indicators: the revised model (1) names the shape (short gaps cluster plus long gaps to the right), (2) states that the mean of 400 m is true but incomplete, (3) mentions the long stretch G9-G14 and the 20-gap sample from Thika Road, and (4) states the frequency density rule. Use a 3-point check (missing, partial, complete) on each group's poster. Ask one learner to explain what density 0.5 for 400 to <1000 means.

<i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings		group.		
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D. TEACHER REFLECTION

After the lesson, reflect: Did learners see the empty 400 to <600 bar and realise that few gaps are near the mean of 400 m? Which groups divided by the class width in metres instead of units of 100 m? Did the wide class genuinely create cognitive conflict, and did frequency density make sense to learners as 'squeezing' a wide class to a fair height? Which learners can read the histogram but cannot yet link it to the pedestrians and the near-miss log? Did the officer remain respected as correct but incomplete? What extra graph paper or pre-labelled axes are needed for Lesson 7?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We drew a histogram of the 20 gaps on the 8,000 m stretch of Thika Road with equal 200 m classes (frequencies 5, 9, 0, 2, 1, 3). Then we merged three classes into 400 to <1000 and found that bar height by frequency was misleading. Frequency density gave 2.5, 4.5, 0.5 and 1.5.
What did I learn?	A histogram has touching bars on a continuous axis, and bar area represents frequency. Frequency density = frequency / class width (in units of 100 m). With unequal class widths, the tallest bar is the one with the highest frequency density, not the highest frequency.
How does this explain the phenomenon?	The histogram shows that most gaps are short (14 of 20 are 300 m or shorter), while a few long gaps of 600 m to 1,100 m lie to the right. These long gaps pull the mean up to 400 m and sit in the long stretch G9-G14 near the school gate. So the officer's mean is true but incomplete, and a graph shows where pedestrians are at risk.

LESSON 7 (80 minutes (2 x 40-minute periods)): Frequency Polygons and Comparing Roads: Same Mean, Different Shape

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners draw frequency polygons by hand and with a spreadsheet or digital device, then compare two polygons (Thika Road and Mombasa Road, Embakasi stretch) to show that similar averages can hide different spread, and use the comparison to advance the driving question.
Knowledge	By the end of the lesson the learner should be able to: (1) plot (midpoint, frequency) points from a grouped table and join them with straight lines; (2) close the polygon on the horizontal axis at both ends; (3) interpret the polygon (modal class, humps, tails, spread); (4) compare two polygons from different roads and make a conclusion supported by data. KICD outcomes: d) represent data using histograms and frequency polygons; e) interpret data from histograms and frequency polygons.
Skills	Plotting points on a scale; using midpoints from the grouped table; using a spreadsheet to build a scatter or line chart; comparing graphs. NGSS-aligned practices: Analyzing and Interpreting Data; Using Mathematics and Computational Thinking; Developing and Using Models; Engaging in Argument from Evidence. KICD core competencies: Critical thinking and problem solving (uses frequency polygons to examine data and make conclusions); Digital literacy (uses digital devices to draw frequency polygons); Communication and collaboration.
Attitudes	KICD values: Integrity (plots and records data accurately); Unity (works in groups using collected data to draw frequency polygons). PCIs: Life Skills (Decision Making), as learners make informed decisions from the polygons; Road safety, as learners discuss where pedestrians are at risk. Learners respect Ms Halima Abdi's mean as correct but incomplete.
Key Inquiry Question	KICD key inquiry questions: 1. What is statistics? 2. How do we represent data? 3. How do we use statistics in day-to-day life? This lesson focuses on question 2 (How do we represent data?) and moves towards question 3.
Purpose in Storyline	Lesson 4 gave us the classes, Lesson 5 the midpoints and Lesson 6 the skill of reading shape from graphs. Lesson 7 turns the same grouped table into a frequency polygon so that two roads can be compared on one set of axes. It shows that the Thika Road mean (grouped estimate 440 m) and the Mombasa Road mean (380 m) are similar, yet the shapes differ: Thika Road has a second small rise at 1000 to <1200 m. This gives the evidence for Lesson 8, where learners make recommendations.
Safety Notes	Learners work indoors with graph paper, rulers and sharp pencils; keep pencils and compasses stored when not in use. Phones and school computers: charge in advance, keep cables out of walkways, and share devices fairly. No roadside work in this lesson. If any learner refers to the real Thika Road, remind the class that the data come from a morning count taken with adult supervision and that learners must never cross at undesignated spots themselves.

B. LESSON OVERVIEW

Back at Ruaraka Hills Secondary School on Thika Road, Mwalimu Wanjiru reminds the class that Ms Halima Abdi's mean of 400 m is correct but incomplete, and that Mzee Otieno's log places 9 of 12 near-misses in one stretch. Learners first predict what a line graph of the gap data (20 gaps, Thika Road, classes of width 200 m) will look like and whether two roads with similar means will look similar. They then draw the Thika Road frequency polygon by hand from the midpoints (100, 300, 500, 700, 900, 1100) and frequencies (5, 9, 0, 2, 1, 3), closing it at (-100, 0) and (1300, 0). Groups repeat the process with a spreadsheet and add the Mombasa Road Embakasi data (4, 8, 5, 2, 1, 0; grouped mean 380 m) on the same axes. In the Explain phase they compare hump, tail and spread, and conclude that similar means (440 m and 380 m) hide different shapes: Thika Road has a second small rise at 1000 to <1200 m, while Mombasa Road has no gap of 1000 m or more. The class updates the DQB and the cumulative model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Frequency Polygons Principles Source: CK-12  Search ARES for similar videos  EXERCISE: Comparing data distributions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the officer's claim and Mzee Otieno's remark. Individually, in 3 minutes, they sketch what they think a line graph of the Thika Road gap data would look like (20 gaps in 6 classes of width 200 m) and write a prediction: 'Mombasa Road has a grouped mean of 380 m and Thika Road 440 m. Will their graphs look alike? Why or why not?' They share in pairs, then with the class. Sketches are kept for comparison at the end of the lesson.	Say: 'Ms Halima Abdi's 400 m is correct but incomplete. Today we draw a picture that shows what the average hides.' Display the grouped table (classes 0 to <200 up to 1000 to <1200; frequencies 5, 9, 0, 2, 1, 3). Ask: 'Without calculating anything, which class do you expect to have the highest point on a line graph, and why?' WAIT TIME 10-15 seconds. Cold-call 4 learners, including at least 2 who have not spoken today, with 'Achieng, what did your partner predict?' Do not correct predictions yet; write 3 different predictions on the board. Ask: 'If two roads have similar means, should their graphs be similar? Raise a hand for yes, no, not sure.' Record the counts.	Predict-Observe-Explain (prediction stage): committing to a prediction in writing exposes learners' ideas that the mean describes the whole data set, so the later comparison of polygons creates a genuine conflict to resolve.	Observe whether sketches show a peak at the 200 to <400 class and whether learners expect the two roads to look similar. Tally the yes/no/not sure counts. Look for the misconception that equal means imply equal graphs. Note learners who mention the long gaps or the school gate unprompted.
Observe Phase  VIDEO: Shapes of distributions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Constructing a Frequency Polygon (PLIX) Source: CK-12  Search ARES for similar readings	Part A, by hand (about 25 minutes): in groups of 4 with graph paper, learners use the Thika Road grouped table from earlier lessons. Kiprop-style table-keepers write the midpoints (100, 300, 500, 700, 900, 1100) next to the frequencies (5, 9, 0, 2, 1, 3). Njeri-style graphers choose scales (horizontal axis from -100 to 1300 m, vertical axis 0 to 10), plot the six points, then add (-100, 0) and (1300, 0) and join all eight points with straight lines using a ruler. Part B, with a spreadsheet or phone (about 20 minutes): groups enter midpoints in one column and frequencies in another, insert a line or scatter-with-lines chart, and add a second column of Mombasa Road frequencies (4, 8, 5, 2, 1, 0) so both polygons appear on the same axes. They label axes ('Gap	Model the first two points on the board: 'The midpoint of 0 to <200 is $(0 + 200)/2 = 100$, so our first point is (100, 5).' Circulate and ask: 'Why do we plot at the midpoint and not at the class boundary?' WAIT TIME 10-15 seconds before taking answers. Ask: 'Why do we close the polygon at -100 and 1300?' and cold-call 3 groups. Prompt with 'What is the frequency of the class 400 to <600 on Thika Road? So where does the line go?' (the line drops to the axis at (500, 0)). Check scales: 'Does your 1100 m point fit on your axis?' For the spreadsheet, show the steps on the teacher device and let group leaders lead. Ask groups: 'Do both polygons start and end on the axis?' Support devices-sharing: rotate who types every 5 minutes. Remind: 'Say	Data Representation and Modelling: translating a table into a graph by hand and by spreadsheet lets learners see how the same numbers produce shape, and checking the spreadsheet against the hand-drawn polygon builds trust in both methods (Using Mathematics and Computational Thinking).	Check each group's polygon: eight points on Thika Road ((-100,0), (100,5), (300,9), (500,0), (700,2), (900,1), (1100,3), (1300,0)); points at midpoints, not boundaries; straight lines with a ruler; closed at both ends; labelled axes. For the Mombasa polygon the points are (100,4), (300,8), (500,5), (700,2), (900,1), (1100,0). Note groups that join the last point to the axis incorrectly, or that miss the zero-frequency class. Ask 2 learners to explain the spreadsheet chart. Record how many groups verify 380 m correctly.

	between crossing points (m)' and 'Frequency'), give a title, and check that both polygons are closed at -100 and 1300. Learners also confirm the Mombasa Road grouped mean: $400 + 2400 + 2500 + 1400 + 900 + 0 = 7,600$, and $7,600/20 = 380$ m. Groups without devices draw both polygons on one sheet using two colours.	Thika Road or Mombasa Road, the unit in metres and the 20-gap sample when you quote any figure.'		
Explain Phase  VIDEO: Example: Comparing distributions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Frequency Polygons (Flexbook) Source: CK-12  Search ARES for similar readings	Groups place the two polygons side by side or on one chart and complete a comparison frame: (1) Where is the highest point on each road? (2) How many classes have gaps of 600 m or more, and how many gaps? (3) Describe the right-hand end of each graph. (4) What do the means 440 m (Thika Road, grouped estimate) and 380 m (Mombasa Road) tell us, and what do they hide? Learners notice that both have a modal class of 200 to <400, but Thika Road has a dip to 0 at 400 to <600, a small rise at 1000 to <1200 (3 gaps), and reaches a longer tail, while Mombasa Road is a single hump with a short right tail, its longest gap in 800 to <1000, and no gap of 1000 m or more. They write one sentence: 'The officer's claim is true but incomplete because ...', using the polygon, and link it to the long stretch G9-G14 (5,250 m, 140 of 182 pedestrians, 9 of 12 near-misses). They then compare their Predict-phase sketches with the real polygons and revise their ideas.	Say: 'Put the two polygons next to each other. Silently look for 1 minute and write down 2 differences and 1 similarity.' WAIT TIME 10-15 seconds after each question. Ask: 'Both roads have the same modal class, 200 to <400. What is different?' Cold-call 5 groups and push for evidence: 'Which points on the graph show that?' Ask: 'Thika Road and Mombasa Road have similar means, 440 m and 380 m. If you were a pedestrian, would you feel equally safe on both? Use the polygons to argue.' Take 2 competing arguments before revealing a view. Press: 'Where on the Thika Road polygon is the school gate stretch?' (the right-hand part, from 600 m upwards). Add: 'A polygon shows shape, but remember this is one weekday morning on 20 gaps. How should we describe our confidence?' Reinforce the language: 'The officer's claim is true but incomplete.' Never say 'wrong'. Remind learners that grouped means are estimates (the true Thika Road mean is 400 m).	Comparing Models / Argument from Evidence (Claim-Evidence-Reasoning): learners make a claim about the two roads, support it with specific points on the polygons, and reason about why the means hide the difference in spread. Comparison forces attention to shape, the idea the mean cannot show.	Listen for statements that use evidence such as 'Thika Road has 3 gaps in 1000 to <1200 but Mombasa Road has none', and for the correct idea that similar means can hide different spread. Check the sentence frame for the phrases 'true but incomplete' and 'estimate'. Mark learners who still say the graphs should look the same because the means are close. Note use of units (m) and the 20-gap sample.
Driving Question Board (DQB) Creation  VIDEO:	Learners return to the Driving Question Board started in Lesson 1. Individually they write on sticky notes: (a) one answer they can now give, with evidence from	Say: 'Go to our DQB. Which question can we now answer using today's polygons?' WAIT TIME 10-15 seconds. Cold-call 4 learners to place a note and say why: 'Njeri,	Driving Question Board: sorting questions under the three KICD key inquiry questions makes progress visible and keeps the investigation learner-driven, linking	Check that notes include evidence from the polygons (not just opinions), that they are placed under the correct KICD question, and that at least some questions

Frequency Polygons Principles Source: CK-12 Search ARES for similar videos EXERCISE: Frequency Polygons Practice Source: CK-12 Search ARES for similar readings	today's polygons; (b) one new question. Examples: 'Why does Mombasa Road have no gap of 1000 m or more?' 'Would the polygon change on a Saturday?' 'How can a polygon help the county decide where to put a footbridge?' Notes are placed under the three KICD questions: What is statistics? How do we represent data? How do we use statistics in day-to-day life? The class moves the questions about drawing frequency polygons and comparing roads to 'answered', and keeps those about recommendations for Lesson 8.	which question is this, and what is your evidence?' Ask: 'Which of our new questions can only be answered by taking more data, for example repeating the count on another day?' Highlight any question about what to recommend and park it for Lesson 8. Praise respectful questions about the officer: 'Good statisticians question the data, not the person.'	today's polygons to the larger puzzle of how an average can say the road is safe while the data show danger.	go beyond today (repeat counts, recommendations). Note learners who write only 'polygons show shape' and prompt them to say what the shape shows about the road.
Model Building Phase VIDEO: Shapes of distributions Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Comparing center and spread Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners update the class model of the phenomenon, a diagram or concept map with the officer's claim at the centre. They add: (1) the frequency polygon shape for Thika Road with its second rise at 1000 to <1200 m; (2) the Mombasa Road polygon as a comparison with a shorter tail; (3) a new link: 'similar means (440 m and 380 m) can hide different spread, so we need to look at the polygon as well as the average'; (4) a connection to the long stretch G9-G14 near the school gate where 140 of 182 pedestrians and 9 of 12 near-misses occurred. Each group adds one arrow and one sentence and notes one limitation (one-morning sample, grouped estimates).	Say: 'Our model needs a new piece today. What does the polygon add that the mean could not show?' WAIT TIME 10-15 seconds. Cold-call 3 groups for their added arrow and sentence. Ask: 'How does this connect to Mzee Otieno's log of 12 near-misses, with 9 in the long stretch?' Ask: 'What would you want to check on the polygon before telling the Parents Association or the county roads office?' Close with an exit slip, to be answered individually in 3 minutes: 'In two sentences, explain why the Thika Road and Mombasa Road means are similar but the polygons are not. Give one number from each road.' Collect the slips. Preview Lesson 8: 'Next time we put everything together and make recommendations.'	Cumulative Model Revision: adding the polygon layer shows learners how each representation (table, mean, histogram, polygon) contributes a different part of the explanation, and the exit slip makes each learner's reasoning visible.	Exit slip indicators: correctly states that both means are similar (440 m and 380 m, grouped estimates) while Thika Road has a second rise at 1000 to <1200 m or a longer right tail; uses numbers and units; uses 'true but incomplete' language; mentions that the sample is one morning. Sort slips into three piles (secure, nearly there, needs reteaching of midpoints or closing the polygon) to plan Lesson 8 groupings.

D. TEACHER REFLECTION

After the lesson ask yourself: (1) Did every group close the polygon at -100 and 1300 and plot at midpoints, and which groups confused boundaries with midpoints? (2) Did learners compare shape (humps, tails, spread) or only compare means? (3) Did the spreadsheet help or distract, and did every learner get a turn at the device? (4)

Which learners still believe that equal means imply equal data? (5) Did learners keep to 'true but incomplete' language about the officer's claim? (6) What will you adjust in Lesson 8 for learners who struggled with scales or with using evidence in a claim? Note names and plan small support groups.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We drew the Thika Road frequency polygon (20 gaps, classes of width 200 m) by plotting (100,5), (300,9), (500,0), (700,2), (900,1), (1100,3), closed at (-100,0) and (1300,0), and drew the Mombasa Road (Embakasi) polygon with frequencies 4, 8, 5, 2, 1, 0 on the same axes, by hand and with a spreadsheet.
What did I learn?	A frequency polygon plots (midpoint, frequency), joins the points with straight lines and is closed on the axis at both ends. Putting two polygons on the same axes lets us compare their shape and spread. The grouped means are similar (Thika Road 440 m, Mombasa Road 380 m) and both have the modal class 200 to <400.
How does this explain the phenomenon?	The officer's claim is true but incomplete. Similar averages can hide different shapes: Thika Road has a second small rise at 1000 to <1200 m and long gaps, while Mombasa Road is a single hump with no gap of 1000 m or more. One average cannot show where pedestrians are at risk, so we use the polygon together with the median, mode and the long stretch G9-G14. This is one morning's sample and should be repeated.

LESSON 8 (80 minutes (double lesson)): Interpreting, Recommending and Presenting Findings: Where Is the Road Really Unsafe?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners bring together every measure, table and graph from Lessons 1-7 to give a final explanation of how the officer's mean of 400 m can be correct while pedestrians are at risk in one stretch of Thika Road, make evidence-based recommendations, and present them to the community.
Knowledge	By the end of the lesson the learner should be able to: (e) interpret data from histograms and frequency polygons; (f) promote data collection, organisation and use for informed decision making. Learners also use the Thika Road 20-gap sample (mean 400 m, median 250 m, mode 200 m, range 1,000 m; grouped frequencies 5, 9, 0, 2, 1, 3; modal class 200 to <400) and the Mombasa Road comparison (frequencies 4, 8, 5, 2, 1, 0; grouped mean 380 m) to state that the officer's claim is true but incomplete.
Skills	Critical thinking and problem solving: use histograms and frequency polygons to examine data and make conclusions. Communication and collaboration: present and share findings with school administrators, parents and local community leaders. NGSS-aligned practices: Analyzing and Interpreting Data; Engaging in Argument from Evidence; Developing and Using Models; Obtaining, Evaluating and Communicating Information.
Attitudes	Values: Integrity (report the data accurately, including its limits as a one-morning sample); Unity (work as a team to build one shared presentation); Love (listen to and consider the audience and each other). PCIs: Life Skills (Decision Making) and Parental Engagement and Empowerment. Road safety is the context throughout.
Key Inquiry Question	1. What is statistics? 2. How do we represent data? 3. How do we use statistics in day-to-day life?
Purpose in Storyline	This is the final lesson. Lessons 1-7 gave learners the claim and the plan (L1), tallies and frequency tables (L2), mean, mode and median (L3), grouped tables (L4), grouped mean and median (L5), histograms and frequency density (L6), and frequency polygons and the road comparison (L7). Lesson 8 does not re-teach these. Learners use them together to answer the driving question, compare their Lesson 1 model with the final model, complete the DQB, and communicate findings to the principal, the Parents Association, the area chief and the county roads representative.
Safety Notes	Rehearsals are done in the classroom only. Do not send learners to Thika Road to repeat counts without adult supervision, a school-approved plan and parental consent. Any follow-up counting should be done from a safe spot such as a footbridge or school gate with a teacher present, never at the roadside. Present Ms Halima Abdi's claim respectfully: her mean of 400 m is correct but incomplete. Present Mzee Otieno's log as supporting evidence, not as blame.

B. LESSON OVERVIEW

Learners return to the puzzle that started the unit: Ms Halima Abdi's mean of about 400 m is correct (8,000 m / 20 gaps), yet children dash across Thika Road near Ruaraka Hills Secondary School and Mzee Otieno says most near-misses happen in one stretch. In the Predict phase, learners decide what they would tell the county roads representative if they could show only one number, then only one graph. In the Observe phase, groups examine the full evidence set at stations: the table of gaps and pedestrian counts, the histogram and polygon of the Thika Road gaps, the Mombasa Road polygon, and Mzee Otieno's near-miss log. They find that 14 of the 20 gaps are 300 m or shorter (median 250 m, mode 200 m), while six gaps of 600 m to 1,100 m (G9-G14) form a 5,250 m long stretch that includes the school gate and holds 140 of 182 undesignated crossers (77%) and 9 of 12 near-misses (75%). In the Explain phase, learners write the Final Explanation with a claim, evidence and reasoning, and draft recommendations. In the DQB phase, they review every question and the three KICD key inquiry questions. In the Model Building phase, they

compare the Lesson 1 model with the final model and rehearse and deliver the community presentation. They end by noting that this is a one-morning sample (Tuesday, 6:30-7:30 am) and should be repeated.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the officer's statement on the board: 'The average distance between crossing points is about 400 m, so the road is well served.' Individually in their books they answer: (1) If you could show the roads office only ONE number from our Thika Road data, which would you choose and why? (2) If you could show only ONE graph, which would it be and what would you want people to see? They then pair up and compare. Learners also predict: 'Where on the 8,000 m would we place new crossing points, and what evidence will persuade the Parents Association and the area chief?' Predictions are recorded on sticky notes under the headings 'Number' and 'Graph'.	Write the officer's claim and the driving question on the board. Say: 'Ms Halima Abdi is not careless. Her 400 m is correct. Today we decide what the whole story is and how to tell it to people who can act.' Give 2 minutes of silent writing. Then say: 'Turn to your partner. Which number, which graph? Convince each other.' Circulate and listen for learners who choose mean only or median only. WAIT TIME: after asking 'Which single number would be fairest to the children crossing near our school gate?', wait 10-15 seconds before taking answers. Cold-call 4 learners from different groups for their choice and reason and write them on the board without judging. Ask: 'What would you lose by showing only one number?' Do not correct yet. Tell learners that their predictions will be revisited at the end.	Commitment to a prediction (Predict-Observe-Explain). Asking learners to choose one number and one graph forces them to commit to what they think statistics can show. Later evidence will confirm or challenge these choices, which makes the final explanation personal because they compare it with their own earlier thinking.	Listen for and record: (a) learners who say the mean 'proves' the road is safe, or who reject the officer's claim as 'wrong' (misconception to address: the claim is true but incomplete); (b) learners who mention spread, a stretch, or a graph shape (ready for interpretation); (c) learners who can name at least two of mean, median, mode in their reasoning. Tally on a class sheet how many chose mean, median, mode, histogram or polygon. Check that all 4 cold-called learners give a reason, not only a choice.
Observe Phase  VIDEO: Interpreting a histogram Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Estimating mean and median in	Groups of 4 (roles from the project: Achieng-style data collector, Kiprop-style calculator, Njeri-style grapher, and a recorder) rotate through four stations, about 6 minutes each, using the Thika Road 20-gap sample. Station 1, Numbers: the sorted list of gaps (100,100,100,150,150,200,200,200,200,250,250,250,300,300,600,700,800,1000,1050,1100) with mean 400 m, median 250 m, mode 200 m and range 1,000 m.	Prepare the four stations beforehand with printed or chalk-drawn data and graphs, calculators and rulers. Say: 'At every station ask: what does this show, and what does it hide?' Circulate. At Station 1 ask: 'The mean is 400 m, yet 14 of 20 gaps are 300 m or shorter. How can that be?' WAIT TIME: wait 10-15 seconds, and do not give the answer. At Station 2 ask: 'Point to where 400 m falls on the	Gallery-style evidence rotation with 'show and hide' analysis. Each station presents one lens (numbers, graphs, people, comparison), and learners ask what each shows and hides. This builds the central idea that single measures summarise but also conceal, and that tables, graphs and measures together give a fuller picture.	Check each group's recording sheet for: (a) accurate figures (5,250 m; 140 of 182; 9 of 12; median 250 m; mean 400 m); (b) at least one statement linking a graph shape to the data (for example, 'the second rise at 1000 to <1200 shows very long gaps'); (c) correct reading of the histogram: bars touch, class 200 to <400 is the modal class; (d) a comparison of the two polygons that goes beyond the mean.

data displays Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners count that 14 of 20 gaps are 300 m or shorter and that only 6 gaps are longer, all over 500 m. Station 2, Graphs: the histogram (classes of width 200 m: 5, 9, 0, 2, 1, 3) and the polygon through (100,5), (300,9), (500,0), (700,2), (900,1), (1100,3) closed at -100 and 1300. They describe the tall bars at 0 to <400, the empty class 400 to <600 and the right tail up to 1000 to <1200, and they locate the mean (400 m) in relation to the bulk of the data. Station 3, Where the people are: gaps in order with pedestrian counts. Learners add G9-G14: $700+1000+1100+800+1050+600 = 5,250$ m (65.6% of the road) and $18+26+31+22+28+15 = 140$ pedestrians (77% of 182). The other 14 gaps total 2,750 m and 42 pedestrians. They also mark the school gate at 4,000 m inside this stretch. Station 4, Near-miss log and comparison: Mzee Otieno's log (12 near-misses, 9 in the long stretch) and the Mombasa Road polygon (4, 8, 5, 2, 1, 0; grouped mean 380 m; longest gap in 800 to <1000; no gap of 1000 m or more). Learners compare the two polygons: similar means (440 vs 380 m) but different spread, with Thika Road showing a second small rise at 1000 to <1200. Each group records three findings on a recording sheet.	histogram. Is that where most gaps are?' At Station 3 ask: 'What percentage of the road is the long stretch, and what percentage of crossers use it?' Support learners who struggle with percentages: $5,250 / 8,000$ and $140 / 182$. At Station 4 ask: 'The two roads have similar means. What do the polygons tell you that the means do not?' Mid-way, cold-call 3 groups to share one surprising finding. Ask the whole class: 'Which station changed your mind the most?' Remind learners to say 'estimate' when quoting grouped means, and to always give the road, the unit and the 20-gap sample.		Observable indicators: learners point to specific bars or points when explaining, and use 'estimate' for grouped values. Note groups that quote only the mean, and prompt: 'What does the median say?'
Explain Phase  VIDEO: Mean, median, & mode example Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Groups write the Final Explanation on a chart using Claim-Evidence-Reasoning. Claim: the officer's mean of 400 m is true but incomplete, because it hides where the risk is. Evidence: at least four pieces from the data (for example 14 of 20 gaps are 300 m or shorter; median 250 m and	Model one CER sentence on the board, then let groups write their own. Say: 'Every recommendation must have a number behind it. If you cannot point to evidence, it is an opinion.' Circulate and challenge: 'Why not put the crossing in a 100 m gap?' (Answer: those gaps already have	Claim-Evidence-Reasoning with peer critique. Writing the Final Explanation requires learners to link several measures and graphs into a single causal argument (long gaps pull up the mean while the risk clusters in one stretch), and peer critique tests whether each recommendation is supported by	Use a four-point check for each chart: (1) the claim states 'true but incomplete'; (2) at least 4 correct data pieces, including both the mean-versus-median contrast and the long-stretch figures; (3) the reasoning explains why a few long gaps raise the mean; (4) each recommendation is tied to

 EXERCISE: Read histograms Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	mode 200 m versus mean 400 m; G9-G14 is 5,250 m, 65.6% of the road, with 140 of 182 pedestrians (77%) and 9 of 12 near-misses (75%); the histogram shows a tail of very long gaps; the Mombasa Road polygon has no gap of 1000 m or more). Reasoning: a few very long gaps pull the mean above the median, so one average cannot show where pedestrians are at risk; median, mode, grouped tables, histograms and polygons together reveal the danger. Next, each group drafts 3 recommendations, such as: extra crossing points (a zebra crossing or footbridge) near the school gate at the 4,000 m mark and within the 1000 to <1200 m gaps; a review of the long stretch G9-G14; and a repeat count on different days and times because this was a one-morning sample (Tuesday, 6:30-7:30 am). Each recommendation must be tied to a piece of evidence. Learners also practise answering a likely question from Ms Halima Abdi, such as: 'Is my average wrong?' with the response: 'Your 400 m is correct, but it does not show the six long gaps.'	many crossings and few pedestrians per gap.) Ask: 'What would you say if the chief asks whether one morning is enough?' WAIT TIME: wait 10-15 seconds, then take responses. Cold-call 3 groups to read their claim aloud; ask the class to signal agree or add-on. Use a gallery walk: groups swap charts and use two stars and a wish to give feedback on evidence and clarity. Stress the Integrity value: report limits honestly. Say: 'Say the claim is true but incomplete, never wrong.' Require the phrase 'one-morning sample' in each explanation.	data. This develops Engaging in Argument from Evidence and Decision Making.	evidence and the sample limitation is stated. Observable indicators: learners correct each other's figures, use the term 'estimate' for grouped values, and refer to specific graph features. Intervene with groups that suggest recommendations with no evidence, or that blame the officer or drivers.
Driving Question Board (DQB) Creation  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners revisit the DQB started in Lesson 1. Groups take turns moving each question into one of three columns: 'Answered by our data', 'Partly answered', and 'New questions'. They write evidence on a sticky note beside each answered question (for example, 'How can an average say the road is safe while the data show danger? - Mean 400 m vs median 250 m; six long gaps G9-G14, 5,250 m'). They then add the three KICD key inquiry questions to the	Say: 'Our board began as a set of puzzles. Let us see what the evidence has done to it.' Read each question aloud and ask: 'Which column, and what is your evidence?' WAIT TIME: allow 10-15 seconds before calling on a group. Cold-call 5 learners to place and justify a question. Push for specifics: 'Which number or graph?' Where a question is only partly answered, ask: 'What data would we need?' (for example, a repeat count on another day or	DQB review with an evidence-linked sort. Sorting questions by how well data answer them helps learners see the whole arc of the unit and recognise the limits of their sample. Linking each answer to the three KICD key inquiry questions ties the project back to the sub-strand.	Observe whether placement of questions is supported by a specific data point, and whether learners can answer the three key inquiry questions in their own words. Look for new questions that are testable with data. Indicator of mastery: learners explain that a question is only partly answered because the sample is one morning on one road, not because the calculations are wrong. Record which key inquiry question is weakest in the class for follow-up.

 HTML: Constructing a Frequency Polygon (PLIX) Source: CK-12  Search ARES for similar readings	board and write a one-line answer under each: (1) What is statistics? Collecting, organising, representing and interpreting data to understand a situation. (2) How do we represent data? Frequency tables, histograms and frequency polygons, each showing a different feature. (3) How do we use statistics in day-to-day life? To check claims and decide where to place crossings, as we did on Thika Road. Finally, each learner writes one new question for the future, such as: 'Do pedestrian counts change on Saturday?' or 'Would a footbridge near the school gate be used?'	other roads such as Mombasa Road or the Kisumu-Kakamega highway). Point out that new questions are a sign of good statistics, not failure. Photograph or copy the board for the presentation and for the class record.		
Model Building Phase  VIDEO: Interpreting a histogram Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Frequency Polygons (Flexbook) Source: CK-12  Search ARES for similar readings	Learners take out the initial model drawn in Lesson 1 (their first explanation of the officer's claim and what they had seen at the school gate). In their groups they redraw a final model on a large sheet showing the 8,000 m of Thika Road as a line with the 21 crossing points, the long stretch G9-G14 shaded (1,550 m to 6,800 m), the school gate at 4,000 m, pedestrian numbers (140 of 182) and near-misses (9 of 12) marked, and the mean (400 m) and median (250 m) labelled, with a small histogram or polygon beside it and arrows showing why the mean is pulled up by the long gaps. Next to it they place the Lesson 1 model and write three sentences: 'What we thought then', 'What changed', and 'What we can now explain'. Groups then use the remaining time to rehearse a 3-minute presentation for the principal, the Parents Association, the area chief and a county roads representative (role-played by volunteers, or by the real guests if present). Each	Hand each group their Lesson 1 model. Say: 'Place your first model next to your final model. What can you explain now that you could not before?' Circulate and prompt: 'Where is the school gate on your model? Why does it matter?' and 'How does your model show why the mean and median differ?' WAIT TIME: wait 10-15 seconds after each question before offering hints. Cold-call 3 groups to describe one change between models. Coach presentations: 'Speak to the audience, not to the chart. Use plain words such as middle value and most common.' Remind them to be respectful: 'Ms Abdi is right about the average; we are adding to it.' Choose one group, or a mixed group, to deliver a full presentation to the guests, and have the rest provide feedback using the rubric. Close with a reflection: 'What is one decision you would now make differently because of data?' Emphasise that the findings come from one morning and should be	Model comparison (Lesson 1 versus final) and communicating to an audience. Placing the first and final models side by side makes visible how learners' thinking grew from 'the average says it is fine' to an evidence-based account of where the risk is. Presenting to real audiences develops Communication and Collaboration and the Life Skill of decision making.	Use the rubric and observation checklist: (a) the final model shows the long stretch, the school gate, the mean-versus-median contrast and the evidence figures correctly; (b) the comparison with the Lesson 1 model names specific changes in thinking; (c) the presentation states 'true but incomplete', uses accurate figures with the road, unit and the 20-gap sample, gives evidence-linked recommendations and notes the one-morning limitation; (d) learners respond appropriately to audience questions. Collect the summary tables and the exit reflection as evidence for the sub-strand rubric: ability to represent and interpret grouped data using histograms and frequency polygons.

	group presents: the claim (true but incomplete), two key graphs, the recommendations and the limits of the sample. Learners give and receive feedback using a rubric (clear claim, evidence, graph reading, recommendation, tone). Finally, they complete the summary table below.	repeated, and that recommendations are for the community to consider.		
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D. TEACHER REFLECTION

After the lesson, consider: Could every group explain why the mean (400 m) differs from the median (250 m) using the long gaps, not just state the numbers? Which learners still described the officer's claim as 'wrong' instead of 'true but incomplete'? Did recommendations come from the data (school gate at 4,000 m, the long stretch G9-G14, the 1000 to <1200 gaps) or from opinion? Did learners acknowledge that this is a one-morning sample (Tuesday, 6:30-7:30 am) and propose a repeat count on other days? Which key inquiry question was weakest on the DQB? Did every learner contribute to the presentation, in line with the values of Unity and Love? Note follow-up needs: learners who need more practice reading histograms and polygons, using percentages, or explaining in plain language. Consider arranging for the group to share the findings with the actual school principal, the Parents Association, the area chief and the county roads office, and to plan a supervised repeat survey.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What we observed: On the 8,000 m stretch of Thika Road (20-gap sample), the mean gap is 400 m but the median is 250 m and the mode is 200 m. 14 of 20 gaps are 300 m or shorter, while six gaps of 600 m to 1,100 m (G9-G14) form a 5,250 m long stretch including the school gate. In this stretch 140 of 182 undesignedated crossers (77%) and 9 of 12 near-misses were recorded. The Mombasa Road polygon has a similar mean (380 m) but no gap of 1000 m or more.
What did I learn?	What we learned: A mean is a useful summary but it can be pulled up by a few very long gaps and hide where people are at risk. Median, mode, grouped tables, histograms and frequency polygons show spread and shape, and together they reveal the danger. Statistics helps us check claims, make informed decisions and communicate with the community. A one-morning sample must be repeated before we are certain.
How does this explain the phenomenon?	What we can now explain: The officer's claim that the average gap is about 400 m is true but incomplete. Because six very long gaps cluster in one stretch near the school gate, the mean looks reassuring while most undesignedated crossings and near-misses occur there. We recommend extra crossing points (for example a zebra crossing or footbridge near the school gate and in the 1000 to <1200 m gaps) and a repeat count, and we presented this evidence to the school administration, parents and community leaders.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.